

NOTES OF MUON OPTIMIZER

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ABSTRACT

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REFERENCES

- Nikita Kornilov, Ohad Shamir, Aleksandr Lobanov, Darina Dvinskikh, Alexander Gasnikov, Innokentiy Shibaev, Eduard Gorbunov, and Samuel Horváth. Accelerated zeroth-order method for non-smooth stochastic convex optimization problem with infinite variance. *Advances in Neural Information Processing Systems*, 36:64083–64102, 2023.
- Zijian Liu and Zhengyuan Zhou. Nonconvex stochastic optimization under heavy-tailed noises: Optimal convergence without gradient clipping. *arXiv preprint arXiv:2412.19529*, 2024.
- Egor Shulgin, Sultan AlRashed, Francesco Orabona, and Peter Richtárik. Beyond the ideal: Analyzing the inexact muon update. *arXiv preprint arXiv:2510.19933*, 2025.
- Yi Wu and Xuejun Wang. The von bahr-esseen type inequality under sub-linear expectations and applications. *arXiv preprint arXiv:2312.08781*, 2023.

A ANALYSIS OF MUON WITH NESTEROV’S MOMENTUM

We are interested in solving the optimization problem

$$\min_{\mathbf{X} \in \mathbb{R}^{d \times d}} f(\mathbf{X})$$

The update rule of the Muon with Nesterov’s momentum is given by,

$$\begin{aligned} \mathbf{C}_k &= \beta \mathbf{C}_{k-1} + \mathbf{G}_k \\ \mathbf{M}_k &= \beta \mathbf{C}_k + \mathbf{G}_k \\ \mathbf{X}_{k+1} &= \mathbf{X}_k - \eta T(\mathbf{M}_k) \end{aligned}$$

where $\beta, \eta > 0$ are the momentum and step size, respectively, and $\mathbf{G}_k$ is an estimate of the gradient $\nabla f(\mathbf{X}_k)$, and $T(\mathbf{M}_k)$ is the estimate of the polar decomposition of $\mathbf{M}_k$. For our analysis, we will make the following assumptions regarding the structure of f and $\mathbf{G}_k$. Note that

$$\begin{aligned} \mathbf{M}_k &= \beta \mathbf{C}_k + \mathbf{G}_k \\ &= \beta(\beta \mathbf{C}_{k-1} + \mathbf{G}_k) + \mathbf{G}_k \\ &= \beta(\mathbf{M}_{k-1} - \mathbf{G}_{k-1} + \mathbf{G}_k) + \mathbf{G}_k \\ &= \beta \mathbf{M}_{k-1} + \mathbf{G}_k + \beta(\mathbf{G}_k - \mathbf{G}_{k-1}). \end{aligned}$$

Assumption A.1. f is L -smooth i.e.

$$f(\mathbf{Y}) \leq f(\mathbf{X}) + \langle \nabla f(\mathbf{X}), \mathbf{Y} - \mathbf{X} \rangle + \frac{L}{2} \|\mathbf{X} - \mathbf{Y}\|_F^2.$$

Assumption A.2. f is μ -strongly convex i.e.

$$f(\mathbf{X}) \geq f(\mathbf{Y}) + \langle \nabla f(\mathbf{Y}), \mathbf{X} - \mathbf{Y} \rangle + \frac{\mu}{2} \|\mathbf{X} - \mathbf{Y}\|_F^2.$$

Assumption A.3. $\mathbf{G}_k$ is an unbiased estimate of $\nabla f(\mathbf{X}_k)$ with bounded variance i.e. $\mathbb{E}[\mathbf{G}_k] = \nabla f(\mathbf{X}_k)$ and

$$\mathbb{E}[\|\mathbf{G}_k - \nabla f(\mathbf{X}_k)\|_F^2 \mid \mathbf{X}_k] \leq \sigma^2.$$

Assumption A.4. $\mathbf{G}_k$ is an unbiased estimate of $\nabla f(\mathbf{X}_k)$ with bounded α -moment i.e. $\mathbb{E}[\mathbf{G}_k] = \nabla f(\mathbf{X}_k)$ and

$$\mathbb{E}[\|\mathbf{G}_k - \nabla f(\mathbf{X}_k)\|_F^\alpha \mid \mathbf{X}_k] \leq \sigma^\alpha$$

with $\alpha \in (1, 2]$

For the approximation error of using the Newton-Schulz or Polar Express, we can make the following assumption similar to [Shulgin et al. \(2025\)](#).

Assumption A.5. There exists a sequence of $0 \leq \Delta_k < 1$ such that

$$\|T(\mathbf{M}_k) - O(\mathbf{M}_k)\|_F \leq \Delta_k.$$

Assumption A.6. There exists a sequence of $0 \leq \Delta < 1$ such that

$$\mathbb{E}[\|T(\mathbf{M}_k) - O(\mathbf{M}_k)\|_F \mid \mathbf{M}_k] \leq \Delta.$$

B ANALYSIS OF MUON WITH NESTEROV’S MOMENTUM AND EXACT POLAR DECOMPOSITION FOR HEAVY TAILED NOISE DISTRIBUTION

Here we consider the batched Muon with Nesterov’s momentum and exact Polar decomposition, i.e.

$$\begin{aligned} \mathbf{G}_k &= \sum_{i=1}^B \mathbf{G}_k^i / B \\ \mathbf{C}_k &= \gamma_k \mathbf{C}_{k-1} + \mathbf{G}_k \\ \mathbf{M}_k &= \beta_k \mathbf{C}_k + \mathbf{G}_k \\ \mathbf{X}_{k+1} &= \mathbf{X}_k - \eta_k O(\mathbf{M}_k) \end{aligned} \quad (1)$$

for some step size $\eta_k > 0$ and momentum parameters $\gamma_k, \beta_k \in (0, 1)$. In the analysis, we will choose $\gamma_k = \beta_{k-1}$ or $\gamma_k = \beta_k$. Moreover, to analyze the heavy-tailed noise distribution, we consider the following assumption on $\mathbf{G}_k^i$

Assumption B.1 (*Generalized Heavy-Tailed Noises*). $\mathbf{G}_k^i$ is an unbiased estimate of $\nabla f(\mathbf{X}_k)$ i.e. $\mathbb{E}[\mathbf{G}_k^i] = \nabla f(\mathbf{X}_k)$ and satisfies

$$\mathbb{E}[\|\mathbf{G}_k^i - \nabla f(\mathbf{X}_k)\|_F^\alpha | \mathbf{X}_k] \leq \sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_k)\|_F^\alpha$$

with $\alpha \in (1, 2]$ for all $i \in [B]$.

Note that we can use the above assumption to get a bound on the noise level of $\mathbf{G}_k$ as follows.

Lemma B.2. Under Assumption B.1, we have

$$\mathbb{E}[\|\mathbf{G}_k - \nabla f(\mathbf{X}_k)\|_F^\alpha | \mathbf{X}_k] \leq \frac{C_\alpha}{B^{\alpha-1}} (\sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_k)\|_F^\alpha) \quad (2)$$

for some $C_\alpha > 0$ independent of batch size B .

Proof. From the definition of $\mathbf{G}_k := \frac{1}{B} \sum_{i=1}^B \mathbf{G}_k^i$, we have

$$\begin{aligned} \mathbb{E}[\|\mathbf{G}_k - \nabla f(\mathbf{X}_k)\|_F^\alpha | \mathbf{X}_k] &= \mathbb{E}\left[\left\|\frac{1}{B} \sum_{i=1}^B \mathbf{G}_k^i - \frac{1}{B} \sum_{i=1}^B \nabla f(\mathbf{X}_k)\right\|_F^\alpha | \mathbf{X}_k\right] \\ &= \frac{1}{B^\alpha} \mathbb{E}\left[\left\|\sum_{i=1}^B (\mathbf{G}_k^i - \nabla f(\mathbf{X}_k))\right\|_F^\alpha | \mathbf{X}_k\right] \\ &\leq \frac{C_\alpha}{B^\alpha} \sum_{i=1}^B \mathbb{E}[\|\mathbf{G}_k^i - \nabla f(\mathbf{X}_k)\|_F^\alpha | \mathbf{X}_k] \end{aligned}$$

where the last inequality follows from the Bahr-Essen inequality (Wu & Wang, 2023) for some constant $C_\alpha > 0$ independent of batch size B (as $\mathbf{G}_k^i - \nabla f(\mathbf{X}_k)$ are mean zero i.i.d. random variables conditioned on $\mathbf{X}_k$). Then, using Assumption B.1, we have

$$\begin{aligned} \mathbb{E}[\|\mathbf{G}_k - \nabla f(\mathbf{X}_k)\|_F^\alpha | \mathbf{X}_k] &\leq \frac{C_\alpha}{B^\alpha} \sum_{i=1}^B (\sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_k)\|_F^\alpha) \\ &= \frac{C_\alpha}{B^{\alpha-1}} (\sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_k)\|_F^\alpha). \end{aligned}$$

Thus, the noise of $\mathbf{G}_k$ decreases as batch size B increases. $\square$

Lemma B.3. Under Assumption A.1, Muon satisfies

$$\eta \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F \leq (f(\mathbf{X}_0) - f_\star) + \frac{Ld}{2} K \eta^2 + \eta(1 + \sqrt{d}) \sum_{k=0}^{K-1} \mathbb{E} \|\epsilon_k\|_F. \quad (3)$$

Proof. Since $\mathbf{O}(\widetilde{\mathbf{M}}_k)$ is orthogonal, $\|\mathbf{O}(\widetilde{\mathbf{M}}_k)\|_F = \sqrt{d}$ and

$$\|\mathbf{X}_{k+1} - \mathbf{X}_k\|_F = \eta \sqrt{d}.$$

By L -smoothness,

$$\begin{aligned} f(\mathbf{X}_{k+1}) &\leq f(\mathbf{X}_k) + \langle \nabla f(\mathbf{X}_k), \mathbf{X}_{k+1} - \mathbf{X}_k \rangle + \frac{L}{2} \|\mathbf{X}_{k+1} - \mathbf{X}_k\|_F^2 \\ &= f(\mathbf{X}_k) - \eta \langle \nabla f(\mathbf{X}_k), \mathbf{O}(\widetilde{\mathbf{M}}_k) \rangle + \frac{Ld}{2} \eta^2 \\ &= f(\mathbf{X}_k) - \eta \langle \widetilde{\mathbf{M}}_k, \mathbf{O}(\widetilde{\mathbf{M}}_k) \rangle + \eta \langle \widetilde{\mathbf{M}}_k - \nabla f(\mathbf{X}_k), \mathbf{O}(\widetilde{\mathbf{M}}_k) \rangle + \frac{Ld}{2} \eta^2 \\ &= f(\mathbf{X}_k) - \eta \|\widetilde{\mathbf{M}}_k\|_* + \eta \langle \epsilon_k, \mathbf{O}(\widetilde{\mathbf{M}}_k) \rangle + \frac{Ld}{2} \eta^2 \\ &\leq f(\mathbf{X}_k) - \eta \|\widetilde{\mathbf{M}}_k\|_F + \eta \|\epsilon_k\|_F \|\mathbf{O}(\widetilde{\mathbf{M}}_k)\|_F + \frac{Ld}{2} \eta^2 \\ &= f(\mathbf{X}_k) - \eta \|\widetilde{\mathbf{M}}_k\|_F + \eta \sqrt{d} \|\epsilon_k\|_F + \frac{Ld}{2} \eta^2. \end{aligned}$$

Using $\|\widetilde{\mathbf{M}}_k\|_F = \|\nabla f(\mathbf{X}_k) + \epsilon_k\|_F \geq \|\nabla f(\mathbf{X}_k)\|_F - \|\epsilon_k\|_F$ gives

$$f(\mathbf{X}_{k+1}) \leq f(\mathbf{X}_k) - \eta \|\nabla f(\mathbf{X}_k)\|_F + \eta(1 + \sqrt{d}) \|\epsilon_k\|_F + \frac{Ld}{2} \eta^2.$$

Taking total expectation and summing from $k = 0$ to $K - 1$ yields

$$\eta \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F \leq (f(\mathbf{X}_0) - f_\star) + \frac{Ld}{2} K \eta^2 + \eta(1 + \sqrt{d}) \sum_{k=0}^{K-1} \mathbb{E} \|\epsilon_k\|_F. \quad (4)$$

□

Lemma B.4.

$$\epsilon_k = \beta^k \epsilon_0 + \beta^2 \sum_{s=1}^k \beta^{k-s} D_s + (1 - \beta) \left((1 + \beta) \xi_k + \sum_{s=1}^{k-1} \beta^{k+1-s} \xi_s - \beta^k \xi_0 \right). \quad (5)$$

Proof. Define rescaled momentum variables

$$\widetilde{\mathbf{C}}_k := (1 - \beta) \mathbf{C}_k, \quad \widetilde{\mathbf{M}}_k := (1 - \beta) \mathbf{M}_k.$$

Then the recursions become

$$\begin{aligned} \widetilde{\mathbf{C}}_k &= (1 - \beta) \mathbf{C}_k = (1 - \beta) (\beta \mathbf{C}_{k-1} + \mathbf{G}_k) = \beta \widetilde{\mathbf{C}}_{k-1} + (1 - \beta) \mathbf{G}_k, \\ \widetilde{\mathbf{M}}_k &= (1 - \beta) \mathbf{M}_k = (1 - \beta) (\beta \mathbf{C}_k + \mathbf{G}_k) = \beta \widetilde{\mathbf{C}}_k + (1 - \beta) \mathbf{G}_k. \end{aligned}$$

For $k \geq 1$, we first express $\widetilde{\mathbf{M}}_k$ in terms of $\widetilde{\mathbf{M}}_{k-1}$:

$$\begin{aligned} \widetilde{\mathbf{M}}_k &= \beta \widetilde{\mathbf{C}}_k + (1 - \beta) \mathbf{G}_k = \beta (\beta \widetilde{\mathbf{C}}_{k-1} + (1 - \beta) \mathbf{G}_k) + (1 - \beta) \mathbf{G}_k \\ &= \beta^2 \widetilde{\mathbf{C}}_{k-1} + (1 - \beta) (1 + \beta) \mathbf{G}_k. \end{aligned}$$

Also $\widetilde{\mathbf{M}}_{k-1} = \beta \widetilde{\mathbf{C}}_{k-1} + (1 - \beta) \mathbf{G}_{k-1}$, so $\beta \widetilde{\mathbf{C}}_{k-1} = \widetilde{\mathbf{M}}_{k-1} - (1 - \beta) \mathbf{G}_{k-1}$ and therefore

$$\widetilde{\mathbf{M}}_k = \beta \widetilde{\mathbf{M}}_{k-1} + (1 - \beta) ((1 + \beta) \mathbf{G}_k - \beta \mathbf{G}_{k-1}).$$

Subtract $\nabla f(\mathbf{X}_k)$ and use $\mathbf{G}_k = \nabla f(\mathbf{X}_k) + \boldsymbol{\xi}_k$ to obtain

$$\begin{aligned}
\boldsymbol{\epsilon}_k &= \widetilde{\mathbf{M}}_k - \nabla f(\mathbf{X}_k) \\
&= \beta \widetilde{\mathbf{M}}_{k-1} + (1-\beta)((1+\beta)\mathbf{G}_k - \beta\mathbf{G}_{k-1}) - \nabla f(\mathbf{X}_k) \\
&= \beta(\boldsymbol{\epsilon}_{k-1} + \nabla f(\mathbf{X}_{k-1})) + (1-\beta)((1+\beta)\mathbf{G}_k - \beta\mathbf{G}_{k-1}) - \nabla f(\mathbf{X}_k) \\
&= \beta\boldsymbol{\epsilon}_{k-1} + (1-\beta)((1+\beta)\boldsymbol{\xi}_k - \beta\boldsymbol{\xi}_{k-1}) + \beta\nabla f(\mathbf{X}_{k-1}) + (1-\beta^2)\nabla f(\mathbf{X}_k) \\
&\quad - (\beta - \beta^2)\nabla f(\mathbf{X}_{k-1}) - \nabla f(\mathbf{X}_k) \\
&= \beta\boldsymbol{\epsilon}_{k-1} + (1-\beta)((1+\beta)\boldsymbol{\xi}_k - \beta\boldsymbol{\xi}_{k-1}) + \beta^2(\nabla f(\mathbf{X}_{k-1}) - \nabla f(\mathbf{X}_k)).
\end{aligned}$$

Hence defining $D_k := \nabla f(\mathbf{X}_{k-1}) - \nabla f(\mathbf{X}_k)$ gives the clean recursion:

$$\boldsymbol{\epsilon}_k = \beta\boldsymbol{\epsilon}_{k-1} + \beta^2 D_k + (1-\beta)((1+\beta)\boldsymbol{\xi}_k - \beta\boldsymbol{\xi}_{k-1}), \quad k \geq 1. \quad (6)$$

Unrolling equation 6 yields (by induction):

$$\boldsymbol{\epsilon}_k = \beta^k \boldsymbol{\epsilon}_0 + \beta^2 \sum_{s=1}^k \beta^{k-s} D_s + (1-\beta) \left((1+\beta)\boldsymbol{\xi}_k + \sum_{s=1}^{k-1} \beta^{k+1-s} \boldsymbol{\xi}_s - \beta^k \boldsymbol{\xi}_0 \right). \quad (7)$$

□

Lemma B.5.

$$\begin{aligned}
\mathbb{E}\|\boldsymbol{\epsilon}_k\|_F &\leq \beta^k \mathbb{E}\|\boldsymbol{\epsilon}_0\|_F + \frac{\beta^2}{1-\beta} L\eta\sqrt{d} + \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1-\beta)^{\frac{\alpha-1}{\alpha}} \\
&\quad + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} (1-\beta) \left((1+\beta) \mathbb{E}\|\nabla f(\mathbf{X}_k)\|_F + \sum_{s=1}^{k-1} \beta^{k+1-s} \mathbb{E}\|\nabla f(\mathbf{X}_s)\|_F + \beta^k \|\nabla f(\mathbf{X}_0)\|_F \right).
\end{aligned} \quad (8)$$

Proof. By L -smoothness,

$$\|D_s\|_F = \|\nabla f(\mathbf{X}_{s-1}) - \nabla f(\mathbf{X}_s)\|_F \leq L\|\mathbf{X}_s - \mathbf{X}_{s-1}\|_F = L\eta\sqrt{d}.$$

Hence from equation ??,

$$\beta^2 \sum_{s=1}^k \beta^{k-s} \mathbb{E}\|D_s\|_F \leq \beta^2 L\eta\sqrt{d} \sum_{j=0}^{k-1} \beta^j \leq \frac{\beta^2}{1-\beta} L\eta\sqrt{d}.$$

View $\mathbb{R}^{d \times d}$ as $\mathbb{R}^{d^2}$ with the Frobenius norm. Because $\mathbb{E}[\boldsymbol{\xi}_s \mid \mathcal{F}_{s-1}] = 0$, the sequence $\{\boldsymbol{\xi}_s\}$ is a martingale difference sequence.

Let the coefficients for the noise combination in equation 5 be defined as:

$$a_0 := -(1-\beta)\beta^k, \quad a_s := (1-\beta)\beta^{k+1-s} \quad (1 \leq s \leq k-1), \quad a_k := (1-\beta)(1+\beta).$$

Then the noise term equals $\sum_{s=0}^k a_s \boldsymbol{\xi}_s$. Let $\kappa_\alpha := 2\sqrt{\pi}C_\alpha^{1/\alpha}$. Using the vector-valued martingale inequality, we have:

$$\mathbb{E} \left\| \sum_{s=0}^k a_s \boldsymbol{\xi}_s \right\|_F \leq 2\sqrt{\pi} \mathbb{E} \left[\left(\sum_{s=0}^k |a_s|^\alpha \|\boldsymbol{\xi}_s\|_F^\alpha \right)^{1/\alpha} \right].$$

Conditioning on $(\mathbf{X}_0, \dots, \mathbf{X}_k)$ and applying Jensen's inequality (since $x \mapsto x^{1/\alpha}$ is concave for $\alpha > 1$), together with the generalized heavy-tailed bound:

$$\begin{aligned}
\mathbb{E} \left[\left(\sum_{s=0}^k |a_s|^\alpha \|\boldsymbol{\xi}_s\|_F^\alpha \right)^{1/\alpha} \right] &\leq \mathbb{E} \left[\left(\sum_{s=0}^k |a_s|^\alpha \mathbb{E}[\|\boldsymbol{\xi}_s\|_F^\alpha \mid \mathbf{X}_s] \right)^{1/\alpha} \right] \\
&\leq \frac{1}{B^{\frac{\alpha-1}{\alpha}}} \mathbb{E} \left[\left(\sum_{s=0}^k |a_s|^\alpha (\sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_s)\|_F^\alpha) \right)^{1/\alpha} \right] \\
&\leq \frac{1}{B^{\frac{\alpha-1}{\alpha}}} \left[\sigma_0 \left(\sum_{s=0}^k |a_s|^\alpha \right)^{1/\alpha} + \sigma_1 \mathbb{E} \left(\sum_{s=0}^k |a_s|^\alpha \|\nabla f(\mathbf{X}_s)\|_F^\alpha \right)^{1/\alpha} \right],
\end{aligned}$$

where the last step uses the subadditivity $(x + y)^{1/\alpha} \leq x^{1/\alpha} + y^{1/\alpha}$.

For the σ_1 term, we use the fact that $(\sum w_s u_s^\alpha)^{1/\alpha} \leq \sum w_s^{1/\alpha} u_s$ for any positive terms when $\alpha > 1$. Setting $w_s = |a_s|^\alpha$ and $u_s = \|\nabla f(\mathbf{X}_s)\|_F$:

$$\mathbb{E} \left(\sum_{s=0}^k |a_s|^\alpha \|\nabla f(\mathbf{X}_s)\|_F^\alpha \right)^{1/\alpha} \leq \sum_{s=0}^k |a_s| \mathbb{E} \|\nabla f(\mathbf{X}_s)\|_F.$$

Additionally, setting $\Gamma_\alpha := (2^\alpha + 1)^{1/\alpha}$, the deterministic sum of coefficients is bounded by:

$$\left(\sum_{s=0}^k |a_s|^\alpha \right)^{1/\alpha} \leq (1 - \beta) \Gamma_\alpha (1 - \beta^\alpha)^{-1/\alpha} \leq (1 - \beta)^{\frac{\alpha-1}{\alpha}} \Gamma_\alpha.$$

Combining these elements yields the bound:

$$\begin{aligned} \mathbb{E} \left\| \sum_{s=0}^k a_s \xi_s \right\|_F &\leq \frac{\kappa_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \left[\Gamma_\alpha \sigma_0 (1 - \beta)^{\frac{\alpha-1}{\alpha}} \right. \\ &\quad \left. + \sigma_1 (1 - \beta) \left((1 + \beta) \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F + \sum_{s=1}^{k-1} \beta^{k+1-s} \mathbb{E} \|\nabla f(\mathbf{X}_s)\|_F + \beta^k \|\nabla f(\mathbf{X}_0)\|_F \right) \right]. \end{aligned} \quad (9)$$

(C) BOUND $\mathbb{E} \|\epsilon_k\|_F$ AND SUM OVER k . From equation 5 and the triangle inequality, combining the drift bound and equation 9 gives for all k :

$$\begin{aligned} \mathbb{E} \|\epsilon_k\|_F &\leq \beta^k \mathbb{E} \|\epsilon_0\|_F + \frac{\beta^2}{1 - \beta} L \eta \sqrt{d} + \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1 - \beta)^{\frac{\alpha-1}{\alpha}} \\ &\quad + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} (1 - \beta) \left((1 + \beta) \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F + \sum_{s=1}^{k-1} \beta^{k+1-s} \mathbb{E} \|\nabla f(\mathbf{X}_s)\|_F + \beta^k \|\nabla f(\mathbf{X}_0)\|_F \right). \end{aligned} \quad (11)$$

□

Theorem B.6. For $\rho := 1 - (1 + \sqrt{d}) \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}}$ we have

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq \frac{f(\mathbf{X}_0) - f_\star}{\eta K \rho} + \frac{Ld}{2\rho} \eta \\ &\quad + \frac{1 + \sqrt{d}}{\rho} \left[\frac{\mathbb{E} \|\epsilon_0\|_F}{K(1 - \beta)} + \frac{\beta^2}{1 - \beta} L \eta \sqrt{d} + \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1 - \beta)^{\frac{\alpha-1}{\alpha}} + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \frac{\|\nabla f(\mathbf{X}_0)\|_F}{K} \right]. \end{aligned}$$

Proof. We now sum equation 11 for $k = 0, \dots, K - 1$. The first three terms give:

$$\sum_{k=0}^{K-1} \beta^k \mathbb{E} \|\epsilon_0\|_F \leq \frac{\mathbb{E} \|\epsilon_0\|_F}{1 - \beta},$$

$$\sum_{k=0}^{K-1} \frac{\beta^2}{1 - \beta} L \eta \sqrt{d} = K \frac{\beta^2}{1 - \beta} L \eta \sqrt{d},$$

$$\sum_{k=0}^{K-1} \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1 - \beta)^{\frac{\alpha-1}{\alpha}} = K \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1 - \beta)^{\frac{\alpha-1}{\alpha}}.$$

For the σ_1 -dependent gradient-coupled part, define $S := \sum_{k=0}^{K-1} \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F$. By swapping the order of summation for the double sum and evaluating the geometric series, the coefficients on the internal gradients sum exactly to $(1 - \beta^2) + \beta^2 = 1$. Thus, we have the clean upper bound:

$$\sum_{k=0}^{K-1} (1 - \beta) \left((1 + \beta) \mathbb{E} \|\nabla f(\mathbf{X}_k)\|_F + \sum_{s=1}^{k-1} \beta^{k+1-s} \mathbb{E} \|\nabla f(\mathbf{X}_s)\|_F \right) \leq S.$$

The initial gradient term is similarly bounded by $\sum_{k=0}^{K-1} (1 - \beta) \beta^k \|\nabla f(\mathbf{X}_0)\|_F \leq \|\nabla f(\mathbf{X}_0)\|_F$. Therefore, the total error sum is bounded by:

$$\sum_{k=0}^{K-1} \mathbb{E} \|\epsilon_k\|_F \leq \frac{\mathbb{E} \|\epsilon_0\|_F}{1 - \beta} + K \frac{\beta^2}{1 - \beta} L \eta \sqrt{d} + K \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1 - \beta)^{\frac{\alpha-1}{\alpha}} + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} (S + \|\nabla f(\mathbf{X}_0)\|_F). \quad (12)$$

Plugging equation 12 back into the one-step descent summation equation ??:

$$\begin{aligned} \eta S &\leq (f(\mathbf{X}_0) - f_\star) + \frac{Ld}{2} K \eta^2 \\ &\quad + \eta(1 + \sqrt{d}) \left[\frac{\mathbb{E} \|\epsilon_0\|_F}{1 - \beta} + K \frac{\beta^2}{1 - \beta} L \eta \sqrt{d} + K \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1 - \beta)^{\frac{\alpha-1}{\alpha}} \right] \\ &\quad + \eta(1 + \sqrt{d}) \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} (S + \|\nabla f(\mathbf{X}_0)\|_F). \end{aligned}$$

Moving the S -term from the RHS to the LHS yields:

$$\begin{aligned} \eta \rho S &\leq (f(\mathbf{X}_0) - f_\star) + \frac{Ld}{2} K \eta^2 \\ &\quad + \eta(1 + \sqrt{d}) \left[\frac{\mathbb{E} \|\epsilon_0\|_F}{1 - \beta} + K \frac{\beta^2}{1 - \beta} L \eta \sqrt{d} + K \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1 - \beta)^{\frac{\alpha-1}{\alpha}} + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \|\nabla f(\mathbf{X}_0)\|_F \right], \end{aligned}$$

where we define $\rho := 1 - (1 + \sqrt{d}) \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}}$. Dividing by $\eta K \rho$ to isolate the average gradient norm gives:

$$\begin{aligned} \frac{1}{K} S &\leq \frac{f(\mathbf{X}_0) - f_\star}{\eta K \rho} + \frac{Ld}{2\rho} \eta \\ &\quad + \frac{1 + \sqrt{d}}{\rho} \left[\frac{\mathbb{E} \|\epsilon_0\|_F}{K(1 - \beta)} + \frac{\beta^2}{1 - \beta} L \eta \sqrt{d} + \frac{\kappa_\alpha \Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0 (1 - \beta)^{\frac{\alpha-1}{\alpha}} + \frac{\kappa_\alpha \sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \frac{\|\nabla f(\mathbf{X}_0)\|_F}{K} \right]. \end{aligned}$$

Finally, plugging in the horizon-dependent parameters $\beta = 1 - \frac{1}{\sqrt{K}}$ and $\eta = \frac{1}{\sqrt{d} K^{3/4}}$ establishes the identities:

$$\frac{1}{\eta K} = \frac{\sqrt{d}}{K^{1/4}}, \quad \frac{1}{K(1 - \beta)} = \frac{1}{K^{1/2}}, \quad \frac{\eta \sqrt{d}}{1 - \beta} = \frac{1}{K^{1/4}}, \quad (1 - \beta)^{\frac{\alpha-1}{\alpha}} = K^{-\frac{\alpha-1}{2\alpha}}.$$

Substituting these directly provides the exact leading rates claimed in the Theorem. $\square$

B.1 SAMPLE COMPLEXITY

In this subsection, we derive the sample complexity from Theorem B.10. Fix any $\varepsilon > 0$. Then choose

$$B \geq \left(2\kappa_\alpha\sigma_1(1+\sqrt{d})\right)^{\alpha/\alpha-1}$$

to ensure $\rho \geq \frac{1}{2}$. Moreover, choose step size $\eta = \frac{1}{\sqrt{K}}$ to get

$$\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] \leq \mathcal{O}(1/\sqrt{K}) + \mathcal{O}(B^{-(\alpha-1)/\alpha}).$$

Now we want the right hand side to be $\mathcal{O}(\varepsilon)$. Thus, we choose $K = \mathcal{O}(\varepsilon^{-2})$ and $B = \mathcal{O}\left(\varepsilon^{\frac{-\alpha}{\alpha-1}}\right)$ to get each term on the right hand side to be of the order $\mathcal{O}(\varepsilon)$. Therefore, the sample complexity is given by

$$\begin{aligned} K \times B &= \mathcal{O}\left(\varepsilon^{-2-\frac{\alpha}{\alpha-1}}\right) \\ &= \mathcal{O}\left(\varepsilon^{\frac{-3\alpha+2}{\alpha-1}}\right). \end{aligned}$$

Note that for $\alpha = 2$, this sample complexity reduces to $\mathcal{O}(\varepsilon^{-4})$, which is optimal.

B.2 PARAMETERS INDEPENDENT OF α

For $\rho := 1 - (1 + \sqrt{d})^{\frac{\kappa_\alpha\sigma_1}{B^{\frac{\alpha-1}{\alpha}}}}$ we have

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq \frac{f(\mathbf{X}_0) - f_\star}{\eta K \rho} + \frac{Ld}{2\rho} \eta \\ &\quad + \frac{1 + \sqrt{d}}{\rho} \left[\frac{\mathbb{E}\|\epsilon_0\|_F}{K(1-\beta)} + \frac{\beta^2}{1-\beta} L\eta\sqrt{d} + \frac{\kappa_\alpha\Gamma_\alpha}{B^{\frac{\alpha-1}{\alpha}}} \sigma_0(1-\beta)^{\frac{\alpha-1}{\alpha}} \right. \\ &\quad \left. + \frac{\kappa_\alpha\sigma_1}{B^{\frac{\alpha-1}{\alpha}}} \frac{\|\nabla f(\mathbf{X}_0)\|_F}{K} \right]. \end{aligned}$$

Finally, plugging in the horizon-dependent parameters $\beta = 1 - \frac{1}{\sqrt{K}}$ and $\eta = \frac{1}{\sqrt{d}K^{3/4}}$ establishes the identities:

$$\frac{1}{\eta K} = \frac{\sqrt{d}}{K^{1/4}}, \quad \frac{1}{K(1-\beta)} = \frac{1}{K^{1/2}}, \quad \frac{\eta\sqrt{d}}{1-\beta} = \frac{1}{K^{1/4}}, \quad (1-\beta)^{\frac{\alpha-1}{\alpha}} = K^{-\frac{\alpha-1}{2\alpha}}.$$

Thus, we get that the sample complexity is $\mathcal{O}(\varepsilon^{-2\alpha/\alpha-1})$.

B.3 MAIN CONVERGENCE ANALYSIS

Lemma B.7. Under Assumption A.1, Muon satisfies

$$\begin{aligned} \sum_{k=0}^{K-1} \eta_k \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq f(\mathbf{X}_0) - f_* \\ &\quad + \left(1 + \sqrt{d}\right) \sum_{k=0}^{K-1} \eta_k \mathbb{E} [\|\mathbf{M}_k - \nabla f(\mathbf{X}_k)\|_F] + \frac{dL}{2} \sum_{k=0}^{K-1} \eta_k^2. \end{aligned}$$

Proof. From the update rule, we have $\|\mathbf{X}_{k+1} - \mathbf{X}_k\|_F = \eta_k \|\mathbf{O}(\mathbf{M}_k)\|_F = \eta_k \sqrt{d}$. Then, using Assumption A.1, we have

$$\begin{aligned} f(\mathbf{X}_{k+1}) &\leq f(\mathbf{X}_k) + \langle \nabla f(\mathbf{X}_k), \mathbf{X}_{k+1} - \mathbf{X}_k \rangle + \frac{L}{2} \|\mathbf{X}_{k+1} - \mathbf{X}_k\|_F^2 \\ &= f(\mathbf{X}_k) - \eta_k \langle \nabla f(\mathbf{X}_k), \mathbf{O}(\mathbf{M}_k) \rangle + \frac{\eta_k^2 dL}{2} \\ &= f(\mathbf{X}_k) - \eta_k \langle \nabla f(\mathbf{X}_k) - \mathbf{M}_k + \mathbf{M}_k, \mathbf{O}(\mathbf{M}_k) \rangle + \frac{\eta_k^2 dL}{2} \\ &= f(\mathbf{X}_k) - \eta_k \langle \mathbf{M}_k, \mathbf{O}(\mathbf{M}_k) \rangle + \eta_k \langle \mathbf{M}_k - \nabla f(\mathbf{X}_k), \mathbf{O}(\mathbf{M}_k) \rangle + \frac{\eta_k^2 dL}{2} \\ &= f(\mathbf{X}_k) - \eta_k \|\mathbf{M}_k\|_* + \eta_k \langle \mathbf{M}_k - \nabla f(\mathbf{X}_k), \mathbf{O}(\mathbf{M}_k) \rangle + \frac{\eta_k^2 dL}{2} \\ &\leq f(\mathbf{X}_k) - \eta_k \|\mathbf{M}_k\|_F + \eta_k \langle \mathbf{M}_k - \nabla f(\mathbf{X}_k), \mathbf{O}(\mathbf{M}_k) \rangle + \frac{\eta_k^2 dL}{2} \\ &\leq f(\mathbf{X}_k) - \eta_k \|\mathbf{M}_k\|_F + \eta_k \|\mathbf{M}_k - \nabla f(\mathbf{X}_k)\|_F \|\mathbf{O}(\mathbf{M}_k)\|_F + \frac{\eta_k^2 dL}{2} \\ &= f(\mathbf{X}_k) - \eta_k \|\mathbf{M}_k\|_F + \eta_k \sqrt{d} \|\mathbf{M}_k - \nabla f(\mathbf{X}_k)\|_F + \frac{\eta_k^2 dL}{2} \\ &\leq f(\mathbf{X}_k) - \eta_k \|\nabla f(\mathbf{X}_k)\|_F + \eta_k \left(1 + \sqrt{d}\right) \|\mathbf{M}_k - \nabla f(\mathbf{X}_k)\|_F + \frac{\eta_k^2 dL}{2}. \end{aligned}$$

Here, we take the total expectations on both sides and sum for $k = 0, 1, \dots, K-1$ to get

$$\begin{aligned} \mathbb{E} [f(\mathbf{X}_K)] &\leq f(\mathbf{X}_0) - \sum_{k=0}^{K-1} \eta_k \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] \\ &\quad + \left(1 + \sqrt{d}\right) \sum_{k=0}^{K-1} \eta_k \mathbb{E} [\|\mathbf{M}_k - \nabla f(\mathbf{X}_k)\|_F] + \frac{dL}{2} \sum_{k=0}^{K-1} \eta_k^2. \end{aligned}$$

Thus, rearranging the terms and using $\mathbb{E} [f(\mathbf{X}_K)] \geq f_*$, we have

$$\begin{aligned} \sum_{k=0}^{K-1} \eta_k \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq f(\mathbf{X}_0) - f_* \\ &\quad + \left(1 + \sqrt{d}\right) \sum_{k=0}^{K-1} \eta_k \mathbb{E} [\|\mathbf{M}_k - \nabla f(\mathbf{X}_k)\|_F] + \frac{dL}{2} \sum_{k=0}^{K-1} \eta_k^2. \end{aligned}$$

This completes the proof of the Lemma. $\square$

Lemma B.8. Muon (Algorithm 1) with $\beta_k = \gamma_k = \beta \in (0, 1)$ satisfies

$$\mathbf{S}_k = \beta^k \nabla f(\mathbf{X}_0) + \sum_{i=0}^k \beta^{k-i+1} \nabla f(\mathbf{X}_i) + \sum_{i=0}^{k-1} \beta^{k+1-i} \boldsymbol{\xi}_i + (1 + \beta) \boldsymbol{\xi}_k, \quad (13)$$

where

$$\begin{aligned} \mathbf{S}_k &:= \mathbf{M}_k - \nabla f(\mathbf{X}_k), \\ \boldsymbol{\xi}_k &:= \mathbf{G}_k - \nabla f(\mathbf{X}_k). \end{aligned}$$

Proof. For simplicity, let us denote

$$\begin{aligned} \mathbf{S}_k &:= \mathbf{M}_k - \nabla f(\mathbf{X}_k), \\ \boldsymbol{\xi}_k &:= \mathbf{G}_k - \nabla f(\mathbf{X}_k). \end{aligned}$$

Then, we can expand $\mathbf{S}_k$ as follows

$$\begin{aligned} \mathbf{S}_k &= \mathbf{M}_k - \nabla f(\mathbf{X}_k) \\ &= \beta_k \mathbf{C}_k + \mathbf{G}_k - \nabla f(\mathbf{X}_k) \\ &= \beta_k (\gamma_k \mathbf{C}_{k-1} + \mathbf{G}_k) + \mathbf{G}_k - \nabla f(\mathbf{X}_k) \\ &= \beta_k \left(\frac{\gamma_k}{\beta_{k-1}} (\mathbf{M}_{k-1} - \mathbf{G}_{k-1}) + \mathbf{G}_k \right) + \mathbf{G}_k - \nabla f(\mathbf{X}_k) \\ &= \frac{\beta_k \gamma_k}{\beta_{k-1}} \mathbf{M}_{k-1} - \frac{\beta_k \gamma_k}{\beta_{k-1}} \mathbf{G}_{k-1} + \beta_k \mathbf{G}_k + \mathbf{G}_k - \nabla f(\mathbf{X}_k) \\ &= \frac{\beta_k \gamma_k}{\beta_{k-1}} (\mathbf{M}_{k-1} - \nabla f(\mathbf{X}_{k-1})) + \frac{\beta_k \gamma_k}{\beta_{k-1}} \nabla f(\mathbf{X}_{k-1}) - \frac{\beta_k \gamma_k}{\beta_{k-1}} \mathbf{G}_{k-1} \\ &\quad + \beta_k \mathbf{G}_k + \mathbf{G}_k - \nabla f(\mathbf{X}_k) \end{aligned}$$

If we set $\gamma_k = \beta_{k-1}$, then we have

$$\begin{aligned} \mathbf{S}_k &= \beta_k \mathbf{S}_{k-1} + \beta_k (\nabla f(\mathbf{X}_{k-1}) - \mathbf{G}_{k-1}) + (\mathbf{G}_k - \nabla f(\mathbf{X}_k)) + \beta_k \mathbf{G}_k \\ &= \beta_k \mathbf{S}_{k-1} - \beta_k \boldsymbol{\xi}_{k-1} + \boldsymbol{\xi}_k + \beta_k (\mathbf{G}_k - \nabla f(\mathbf{X}_k)) + \beta_k \nabla f(\mathbf{X}_k) \\ &= \beta_k \mathbf{S}_{k-1} - \beta_k \boldsymbol{\xi}_{k-1} + (1 + \beta_k) \boldsymbol{\xi}_k + \beta_k \nabla f(\mathbf{X}_k). \end{aligned}$$

Here, we set $\beta_k = \beta \in (0, 1)$ to have

$$\mathbf{S}_k = \beta \mathbf{S}_{k-1} - \beta \boldsymbol{\xi}_{k-1} + (1 + \beta) \boldsymbol{\xi}_k + \beta \nabla f(\mathbf{X}_k).$$

Then, unrolling the recurrence relation, we have

$$\begin{aligned} \mathbf{S}_k &= \beta^k \mathbf{S}_0 + \sum_{i=1}^k \beta^{k-i} (\beta \nabla f(\mathbf{X}_i) - \beta \boldsymbol{\xi}_{i-1} + (1 + \beta) \boldsymbol{\xi}_i) \\ &= \beta^k \mathbf{S}_0 + \beta \sum_{i=1}^k \beta^{k-i} \nabla f(\mathbf{X}_i) - \beta^k \boldsymbol{\xi}_0 + (1 + \beta) \boldsymbol{\xi}_k + \sum_{i=1}^{k-1} \beta^{k+1-i} \boldsymbol{\xi}_i. \end{aligned}$$

Now note that $\mathbf{S}_0$ can be expanded as

$$\begin{aligned} \mathbf{S}_0 &= \beta \mathbf{C}_0 + \mathbf{G}_0 \\ &= \beta (\beta \mathbf{C}_{-1} + \mathbf{G}_0) + \mathbf{G}_0 \\ &= (1 + \beta) \mathbf{G}_0 \\ &= (1 + \beta) (\nabla f(\mathbf{X}_0) + \boldsymbol{\xi}_0). \end{aligned}$$

Therefore $\beta^k \mathbf{S}_0 - \beta^k \boldsymbol{\xi}_0 = (1 + \beta) \beta^k \nabla f(\mathbf{X}_0) + \beta^{k+1} \boldsymbol{\xi}_0$, and we have

$$\mathbf{S}_k = \beta^k \nabla f(\mathbf{X}_0) + \sum_{i=0}^k \beta^{k-i+1} \nabla f(\mathbf{X}_i) + (1 + \beta) \boldsymbol{\xi}_k + \sum_{i=0}^{k-1} \beta^{k+1-i} \boldsymbol{\xi}_i.$$

This completes the proof of the Lemma. $\square$

Lemma B.9. Muon with Nesterov's momentum satisfies

$$\begin{aligned} \mathbb{E} [\|\mathbf{S}_k\|_F] &\leq \beta^k \|\nabla f(\mathbf{X}_0)\|_F + \left(1 + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}}\right) \sum_{i=0}^k \beta^{k+1-i} \mathbb{E} [\|\nabla f(\mathbf{X}_i)\|_F] \\ &\quad + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] + \frac{4\sqrt{\pi}C_\alpha\sigma_0}{B^{\alpha-1/\alpha}} \left(\sum_{i=0}^{k+1} \beta^{\alpha(k+1-i)}\right)^{1/\alpha}. \end{aligned}$$

Proof. From (13) and the triangle inequality, we have

$$\begin{aligned} \mathbb{E} [\|\mathbf{S}_k\|_F] &\leq \beta^k \|\nabla f(\mathbf{X}_0)\|_F + \sum_{i=0}^k \beta^{k-i+1} \mathbb{E} [\|\nabla f(\mathbf{X}_i)\|_F] \\ &\quad + \mathbb{E} \left[\left\| (1+\beta)\boldsymbol{\xi}_k + \sum_{i=0}^{k-1} \beta^{k+1-i} \boldsymbol{\xi}_i \right\|_F \right] \end{aligned}$$

after taking the total expectation. Note that we can bound the last term of above inequality as follows

$$\begin{aligned} &\mathbb{E} \left[\left\| (1+\beta)\boldsymbol{\xi}_k + \sum_{i=0}^{k-1} \beta^{k+1-i} \boldsymbol{\xi}_i \right\|_F \right] \\ &\stackrel{(14)}{\leq} 2\sqrt{\pi} \mathbb{E} \left[\left((1+\beta)^\alpha \|\boldsymbol{\xi}_k\|_F^\alpha + \sum_{i=0}^{k-1} \beta^{\alpha(k+1-i)} \|\boldsymbol{\xi}_i\|_F^\alpha \right)^{1/\alpha} \right] \\ &\stackrel{(2)}{\leq} \frac{2\sqrt{\pi}C_\alpha\sigma_0}{B^{\alpha-1/\alpha}} \mathbb{E} \left[\left((1+\beta)^\alpha (\sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_k)\|_F^\alpha) + \sum_{i=0}^{k-1} \beta^{\alpha(k+1-i)} (\sigma_0^\alpha + \sigma_1^\alpha \|\nabla f(\mathbf{X}_i)\|_F^\alpha) \right)^{1/\alpha} \right] \\ &\leq \frac{4\sqrt{\pi}C_\alpha\sigma_0}{B^{\alpha-1/\alpha}} \left(1 + \beta^\alpha + \sum_{i=0}^{k-1} \beta^{\alpha(k+1-i)} \right)^{1/\alpha} + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}} \left((1+\beta) \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] \right. \\ &\quad \left. + \sum_{i=0}^{k-1} \beta^{k+1-i} \mathbb{E} [\|\nabla f(\mathbf{X}_i)\|_F] \right). \end{aligned}$$

Hence, combining the two inequalities, we have

$$\begin{aligned} \mathbb{E} [\|\mathbf{S}_k\|_F] &\leq \beta^k \|\nabla f(\mathbf{X}_0)\|_F + \left(1 + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}}\right) \sum_{i=0}^{k-1} \beta^{k+1-i} \mathbb{E} [\|\nabla f(\mathbf{X}_i)\|_F] \\ &\quad + \left(\beta + (1+\beta) \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}} \right) \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] + \frac{4\sqrt{\pi}C_\alpha\sigma_0}{B^{\alpha-1/\alpha}} \left(\sum_{i=0}^{k+1} \beta^{\alpha(k+1-i)} \right)^{1/\alpha} \\ &= \beta^k \|\nabla f(\mathbf{X}_0)\|_F + \left(1 + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}}\right) \sum_{i=0}^k \beta^{k+1-i} \mathbb{E} [\|\nabla f(\mathbf{X}_i)\|_F] \\ &\quad + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] + \frac{4\sqrt{\pi}C_\alpha\sigma_0}{B^{\alpha-1/\alpha}} \left(\sum_{i=0}^{k+1} \beta^{\alpha(k+1-i)} \right)^{1/\alpha}. \end{aligned}$$

This completes the proof of the Lemma. $\square$

Theorem B.10. Consider Muon with Nesterov's momentum and suppose Assumptions A.1 and B.1 hold. We choose the batch size

$$B > \left(2\sqrt{\pi}C_\alpha\sigma_1(1+\sqrt{d})\right)^{\alpha/1-\alpha}$$

and a momentum parameter

$$0 < \beta < \frac{\left(1 - \frac{2\sqrt{\pi}C_\alpha\sigma_1(1+\sqrt{d})}{B^{\alpha-1/\alpha}}\right)}{(2+\sqrt{d})}.$$

Then we have

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq \frac{f(\mathbf{X}_0) - f_*}{\eta\rho K} + \frac{dL\eta}{2\rho} + \frac{4\sqrt{\pi}(1+\sqrt{d})C_\alpha\sigma_0}{\rho(1-\beta^\alpha)^{1/\alpha}B^{\alpha-1/\alpha}} \\ &\quad + \frac{(1+\sqrt{d})\|\nabla f(\mathbf{X}_0)\|_F}{\rho(1-\beta)K}. \end{aligned}$$

where

$$\rho := 1 - (1+\sqrt{d})\left(\frac{\beta + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}}}{1-\beta}\right) > 0.$$

Proof. Note that, from Lemma B.9 we have

$$\begin{aligned} \sum_{k=0}^{K-1} \mathbb{E} [\|\mathbf{S}_k\|_F] &= \|\nabla f(\mathbf{X}_0)\|_F \sum_{k=0}^{K-1} \beta^k + \left(1 + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}}\right) \sum_{k=0}^{K-1} \sum_{i=0}^k \beta^{k+i-1} \mathbb{E} [\|\nabla f(\mathbf{X}_i)\|_F] \\ &\quad + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] + \frac{4\sqrt{\pi}C_\alpha\sigma_0}{B^{\alpha-1/\alpha}} \sum_{k=0}^{K-1} \left(\sum_{i=0}^{k+1} \beta^{\alpha(k+1-i)}\right)^{1/\alpha} \\ &= \|\nabla f(\mathbf{X}_0)\|_F \sum_{k=0}^{K-1} \beta^k + \left(1 + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}}\right) \sum_{i=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_i)\|_F] \sum_{k=i}^{K-1} \beta^{k+i-1} \\ &\quad + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] + \frac{4\sqrt{\pi}C_\alpha\sigma_0}{B^{\alpha-1/\alpha}} \sum_{k=0}^{K-1} \left(\sum_{i=0}^{k+1} \beta^{\alpha(k+1-i)}\right)^{1/\alpha} \\ &\leq \frac{\|\nabla f(\mathbf{X}_0)\|_F}{1-\beta} + \left(1 + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}}\right) \sum_{i=0}^{K-1} \frac{\beta}{1-\beta} \mathbb{E} [\|\nabla f(\mathbf{X}_i)\|_F] \\ &\quad + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] + \frac{4\sqrt{\pi}C_\alpha\sigma_0}{B^{\alpha-1/\alpha}} \sum_{k=0}^{K-1} (1-\beta^\alpha)^{-1/\alpha} \\ &= \frac{\|\nabla f(\mathbf{X}_0)\|_F}{1-\beta} + \left(1 + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}}\right) \sum_{k=0}^{K-1} \frac{\beta}{1-\beta} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] \\ &\quad + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] + \frac{4\sqrt{\pi}C_\alpha\sigma_0}{B^{\alpha-1/\alpha}} K (1-\beta^\alpha)^{-1/\alpha}. \end{aligned}$$

Thus, with $\eta_k = \eta$ for all k , we have

$$\eta \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] \leq f(\mathbf{X}_0) - f_* + \frac{dLK}{2}\eta^2 + (1+\sqrt{d})\eta \sum_{k=0}^{K-1} \mathbb{E} [\|\mathbf{S}_k\|_F].$$

Thus, using the bound of $\sum_{k=0}^{K-1} \mathbb{E} [\|\mathbf{S}_k\|_F]$ and rearranging the terms, we have

$$\begin{aligned} \eta \sum_{k=0}^{K-1} \left(1 - \left(1 + \sqrt{d} \right) \left(\frac{\beta + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}}}{1 - \beta} \right) \right) \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq f(\mathbf{X}_0) - f_* + \frac{dLK}{2}\eta^2 \\ &\quad + \frac{4\sqrt{\pi} \left(1 + \sqrt{d} \right) C_\alpha \sigma_0 \eta K}{(1 - \beta^\alpha)^{1/\alpha} B^{\alpha-1/\alpha}} \\ &\quad + \frac{\left(1 + \sqrt{d} \right) \eta}{1 - \beta} \|\nabla f(\mathbf{X}_0)\|_F. \end{aligned}$$

For $\sigma_1 > 0$, we choose a large batch size $B > \left(2\sqrt{\pi}C_\alpha\sigma_1 \left(1 + \sqrt{d} \right) \right)^{\alpha/(1-\alpha)}$ and a momentum $0 < \beta < \left(1 - \frac{2\sqrt{\pi}C_\alpha\sigma_1(1+\sqrt{d})}{B^{\alpha-1/\alpha}} \right) / (2+\sqrt{d})$ to ensure

$$\rho := 1 - \left(1 + \sqrt{d} \right) \left(\frac{\beta + \frac{2\sqrt{\pi}C_\alpha\sigma_1}{B^{\alpha-1/\alpha}}}{1 - \beta} \right) > 0.$$

Then we get

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq \frac{f(\mathbf{X}_0) - f_*}{\eta\rho K} + \frac{dL\eta}{2\rho} + \frac{4\sqrt{\pi} \left(1 + \sqrt{d} \right) C_\alpha \sigma_0}{\rho(1 - \beta^\alpha)^{1/\alpha} B^{\alpha-1/\alpha}} \\ &\quad + \frac{\left(1 + \sqrt{d} \right) \|\nabla f(\mathbf{X}_0)\|_F}{\rho(1 - \beta)K}. \end{aligned}$$

This completes the proof of the Theorem. $\square$

Remark B.11. For $\sigma_1 = 0$, we have

$$\begin{aligned} \eta \sum_{k=0}^{K-1} \left(1 - \left(1 + \sqrt{d} \right) \frac{\beta}{1 - \beta} \right) \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq f(\mathbf{X}_0) - f_* + \frac{dLK}{2}\eta^2 \\ &\quad + \frac{4\sqrt{\pi} \left(1 + \sqrt{d} \right) \sigma_0 \eta K}{(1 - \beta^\alpha)^{1/\alpha}} \\ &\quad + \frac{\left(1 + \sqrt{d} \right) \eta}{1 - \beta} \|\nabla f(\mathbf{X}_0)\|_F. \end{aligned}$$

Note that we can choose $\beta < 1/(2+\sqrt{d})$ to ensure

$$\rho := 1 - \left(1 + \sqrt{d} \right) \frac{\beta}{1 - \beta} > 0.$$

Therefore, this choice of $\beta < 1/(2+\sqrt{d})$ implies

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq \frac{f(\mathbf{X}_0) - f_*}{\eta\rho K} + \frac{dL\eta}{2\rho} + \frac{4\sqrt{\pi} \left(1 + \sqrt{d} \right) \sigma_0}{\rho(1 - \beta^\alpha)^{1/\alpha} B^{\alpha-1/\alpha}} \\ &\quad + \frac{\left(1 + \sqrt{d} \right) \|\nabla f(\mathbf{X}_0)\|_F}{\rho(1 - \beta)K}. \end{aligned}$$

Thus, we do not require any restriction on batch size B for $\sigma_1 = 0$.

B.4 SAMPLE COMPLEXITY

In this subsection, we derive the sample complexity from Theorem B.10. Fix any $\varepsilon > 0$. Then choose

$$\begin{aligned} B &\geq \left(16\sqrt{\pi}C_\alpha\sigma_1(1+\sqrt{d})\right)^{\alpha/\alpha-1} \\ \beta &\leq \frac{1}{8+4\sqrt{d}} \end{aligned}$$

to ensure $\rho \geq \frac{1}{2}$. Moreover, choose step size $\eta = \frac{1}{\sqrt{K}}$ to get

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq \frac{2(f(\mathbf{X}_0) - f_*)}{\sqrt{K}} + \frac{dL}{\sqrt{K}} + \frac{8\sqrt{\pi}(1+\sqrt{d})\sigma_0}{(1-\beta^\alpha)^{1/\alpha} B^{\alpha-1/\alpha}} \\ &\quad + \frac{2(1+\sqrt{d})\|\nabla f(\mathbf{X}_0)\|_F}{(1-\beta)K}. \end{aligned}$$

Now we want the right hand side to be $\mathcal{O}(\varepsilon)$. Thus, we choose $K = \mathcal{O}(\varepsilon^{-2})$ and $B = \mathcal{O}\left(\varepsilon^{\frac{-\alpha}{\alpha-1}}\right)$ to get each term on the right hand side to be of the order $\mathcal{O}(\varepsilon)$. Therefore, the sample complexity is given by

$$\begin{aligned} K \times B &= \mathcal{O}\left(\varepsilon^{-2-\frac{\alpha}{\alpha-1}}\right) \\ &= \mathcal{O}\left(\varepsilon^{\frac{-3\alpha+2}{\alpha-1}}\right). \end{aligned}$$

Note that for $\alpha = 2$, this sample complexity reduces to $\mathcal{O}(\varepsilon^{-4})$, which is optimal.

B.5 PARAMETERS INDEPENDENT OF α

In the case of $\sigma_1 = 0$, we can choose $\beta < 1/4+2\sqrt{d}$ and,

$$\rho := 1 - \left(1 + \sqrt{d}\right) \frac{\beta}{1-\beta} > \frac{1}{2}.$$

to get

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq \frac{2(f(\mathbf{X}_0) - f_*)}{\eta K} + dL\eta + \frac{8\sqrt{\pi}(1+\sqrt{d})\sigma_0}{(1-\beta^\alpha)^{1/\alpha} B^{\alpha-1/\alpha}} \\ &\quad + \frac{2(1+\sqrt{d})\|\nabla f(\mathbf{X}_0)\|_F}{(1-\beta)K}. \end{aligned}$$

Then, for batch size $B = 1$, we have

$$\begin{aligned} \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_F] &\leq \frac{2(f(\mathbf{X}_0) - f_*)}{\eta K} + dL\eta + \frac{8\sqrt{\pi}(1+\sqrt{d})\sigma_0}{(1-\beta^\alpha)^{1/\alpha}} \\ &\quad + \frac{2(1+\sqrt{d})\|\nabla f(\mathbf{X}_0)\|_F}{(1-\beta)K}. \end{aligned}$$

Now, we choose $\beta = 1 -$

B.6 TECHNICAL LEMMAS

Lemma B.12. Suppose $\mathbf{W}_t \in \mathbb{R}^{d \times d}$ is a sequence of matrices such that $\forall t \in \mathbb{N}$ we have $\mathbb{E}[\mathbf{W}_t | \mathbf{W}_1, \dots, \mathbf{W}_{t-1}] = \mathbf{0}$. Then for any $p \in (1, 2]$, we have

$$\mathbb{E} \left[\left\| \sum_{t=1}^T \mathbf{W}_t \right\|_F \right] \leq 2\sqrt{\pi} \mathbb{E} \left[\left(\sum_{t=1}^T \|\mathbf{W}_t\|_F^p \right)^{1/p} \right]. \quad (14)$$

Proof. Following Kornilov et al. (2023), consider a random gaussian vector $\mathbf{g} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and define $\mathbf{v}_t := \mathbf{g}^\top \mathbf{W}_t$, where $\mathbf{g}$ is independent of $\mathbf{W}_t$. Moreover, define the sigma algebras $\mathcal{H}_1 := \sigma(\mathbf{v}_{t-1}, \dots, \mathbf{v}_1)$ and $\mathcal{H}_2 := \sigma(\mathbf{W}_{t-1}, \dots, \mathbf{W}_1, \mathbf{g})$. Observe that $\mathcal{H}_1 \subset \mathcal{H}_2$, and by the tower property

$$\begin{aligned} \mathbb{E}[\mathbf{g}^\top \mathbf{W}_t | \mathcal{H}_1] &= \mathbb{E}[\mathbb{E}[\mathbf{g}^\top \mathbf{W}_t | \mathcal{H}_2] | \mathcal{H}_1] \\ &= \mathbb{E}[\mathbf{g}^\top \mathbb{E}[\mathbf{W}_t | \mathcal{H}_2] | \mathcal{H}_1] \\ &= \mathbf{0} \end{aligned}$$

where the second equality holds as $\mathbf{g}$ is $\mathcal{H}_2$ -measurable, while the last equality follows from the independence of $\mathbf{g}$ and $\mathbf{W}_t$, along with the given assumption $\mathbb{E}[\mathbf{W}_t | \mathbf{W}_1, \dots, \mathbf{W}_{t-1}] = \mathbf{0}$. Therefore, from Liu & Zhou, 2024, Lemma 4.3, we have

$$\mathbb{E} \left[\left\| \sum_{t=1}^T \mathbf{v}_t \right\| \right] \leq 2\sqrt{2} \mathbb{E} \left[\left(\sum_{t=1}^T \|\mathbf{v}_t\|^p \right)^{1/p} \right]$$

for any $p \in (1, 2]$. Now the left hand side of the above inequality can be lower bounded as follows

$$\begin{aligned} \mathbb{E} \left[\left\| \sum_{t=1}^T \mathbf{v}_t \right\| \right] &= \mathbb{E} \left[\left\| \sum_{t=1}^T \mathbf{g}^\top \mathbf{W}_t \right\| \right] \\ &= \mathbb{E} \left[\mathbb{E} \left[\left\| \mathbf{g}^\top \left(\sum_{t=1}^T \mathbf{W}_t \right) \right\| \middle| \sum_{t=1}^T \mathbf{W}_t \right] \right] \\ &\stackrel{(17)}{\geq} \sqrt{\frac{2}{\pi}} \mathbb{E} \left[\left\| \sum_{t=1}^T \mathbf{W}_t \right\|_F \right]. \end{aligned}$$

For the right hand side, we have

$$\mathbb{E} \left[\left(\sum_{t=1}^T \|\mathbf{v}_t\|^p \right)^{1/p} \right] = \mathbb{E} \left[\left(\sum_{t=1}^T \|\mathbf{g}^\top \mathbf{W}_t\|^p \right)^{1/p} \right]$$

Conditioning on $\mathbf{W}_1, \mathbf{W}_2, \dots, \mathbf{W}_T$, we get

$$\begin{aligned} \mathbb{E} \left[\left(\sum_{t=1}^T \|\mathbf{g}^\top \mathbf{W}_t\|^p \right)^{1/p} \right] &= \mathbb{E} \left[\mathbb{E} \left[\left(\sum_{t=1}^T \|\mathbf{g}^\top \mathbf{W}_t\|^p \right)^{1/p} \middle| \mathbf{W}_1, \dots, \mathbf{W}_T \right] \right] \\ &\leq \mathbb{E} \left[\left(\mathbb{E} \left[\sum_{t=1}^T \|\mathbf{g}^\top \mathbf{W}_t\|^p \middle| \mathbf{W}_1, \dots, \mathbf{W}_T \right] \right)^{1/p} \right] \\ &= \mathbb{E} \left[\left(\sum_{t=1}^T \mathbb{E} [\|\mathbf{g}^\top \mathbf{W}_t\|^p | \mathbf{W}_t] \right)^{1/p} \right] \end{aligned}$$

where the inequality follows from $p > 1$ and (16). Suppose the singular value decomposition of $\mathbf{W}_t$ is given by $\mathbf{W}_t = \mathbf{U}_t \mathbf{\Sigma}_t \mathbf{V}_t^\top$. Then, defining $\hat{\mathbf{g}}_t = \mathbf{U}_t^\top \mathbf{g} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, we have

$$\begin{aligned} \mathbb{E} [\|\mathbf{g}^\top \mathbf{W}_t\|^p | \mathbf{W}_t] &= \mathbb{E} [\|\mathbf{V}_t^\top \mathbf{\Sigma}_t^\top \hat{\mathbf{g}}_t\|^p | \mathbf{W}_t] \\ &= \mathbb{E} [\|\mathbf{\Sigma}_t^\top \hat{\mathbf{g}}_t\|^p | \mathbf{W}_t] \end{aligned}$$

where the last equality follows from the orthogonality of $\mathbf{V}_t$. Now, denote the singular values of $\mathbf{W}_t$ as $\sigma_{t,1}, \dots, \sigma_{t,r_t}$, where r_t is the rank of $\mathbf{W}_t$, and denote the i -th coordinate of $\hat{\mathbf{g}}_t$ by $\hat{g}_{t,i}$. Then we obtain

$$\begin{aligned} \mathbb{E} \left[\|\mathbf{g}^\top \mathbf{W}_t\|^p \middle| \mathbf{W}_t \right] &= \mathbb{E} \left[\left(\sum_{i=1}^{r_t} \sigma_{t,i}^2 \hat{g}_{t,i}^2 \right)^{p/2} \middle| \mathbf{W}_t \right] \\ &= \left(\sum_{j=1}^{r_t} \sigma_{t,j}^2 \right)^{p/2} \mathbb{E} \left[\left(\sum_{i=1}^{r_t} \frac{\sigma_{t,i}^2}{\sum_{j=1}^{r_t} \sigma_{t,j}^2} \hat{g}_{t,i}^2 \right)^{p/2} \middle| \mathbf{W}_t \right] \\ &\leq \left(\sum_{j=1}^{r_t} \sigma_{t,j}^2 \right)^{p/2} \left(\sum_{i=1}^{r_t} \frac{\sigma_{t,i}^2}{\sum_{j=1}^{r_t} \sigma_{t,j}^2} \mathbb{E} [\hat{g}_{t,i}^2 | \mathbf{W}_t] \right)^{p/2} \end{aligned}$$

where the last inequality follows from $p \leq 2$ and (16). Then, using $\mathbb{E} [\hat{g}_{t,i}^2 | \mathbf{W}_t] = 1$ from $\hat{g}_{t,i} \sim \mathcal{N}(0, 1)$, we have

$$\begin{aligned} \mathbb{E} \left[\|\mathbf{g}^\top \mathbf{W}_t\|^p \middle| \mathbf{W}_t \right] &\leq \left(\sum_{j=1}^{r_t} \sigma_{t,j}^2 \right)^{p/2} \\ &= \|\mathbf{W}_t\|_F^p. \end{aligned}$$

Thus, we obtain

$$\mathbb{E} \left[\left(\sum_{t=1}^T \|\mathbf{v}_t\|^p \right)^{1/p} \right] \leq \mathbb{E} \left[\left(\sum_{t=1}^T \|\mathbf{W}_t\|_F^p \right)^{1/p} \right].$$

Finally, we get

$$\sqrt{\frac{2}{\pi}} \mathbb{E} \left[\left\| \sum_{t=1}^T \mathbf{W}_t \right\|_F \right] \leq 2\sqrt{2} \mathbb{E} \left[\left(\sum_{t=1}^T \|\mathbf{W}_t\|_F^p \right)^{1/p} \right]$$

or equivalently,

$$\mathbb{E} \left[\left\| \sum_{t=1}^T \mathbf{W}_t \right\|_F \right] \leq 2\sqrt{\pi} \mathbb{E} \left[\left(\sum_{t=1}^T \|\mathbf{W}_t\|_F^p \right)^{1/p} \right].$$

This completes the proof of the Lemma. $\square$

Lemma B.13 (*Jensen's Inequality*). For the random variable $\mathbf{X}$, then we have

1. if φ is convex

$$\varphi(\mathbb{E}[\mathbf{X}]) \leq \mathbb{E}[\varphi(\mathbf{X})]. \quad (15)$$

2. and if φ is concave

$$\mathbb{E}[\varphi(\mathbf{X})] \leq \varphi(\mathbb{E}[\mathbf{X}]). \quad (16)$$

Lemma B.14. For a random vector $\mathbf{g} \sim \mathcal{N}_d(\mathbf{0}, \mathbf{I})$ and a matrix $\mathbf{M} \in \mathbb{R}^{d \times d}$, we have

$$\mathbb{E}[\|\mathbf{g}^\top \mathbf{M}\|] \geq \sqrt{\frac{2}{\pi}} \|\mathbf{M}\|_F. \quad (17)$$

Proof. Consider the singular value decomposition of $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ and define $\hat{\mathbf{g}} := \mathbf{U}^\top \mathbf{g}$. As $\mathbf{U}$ is orthogonal, $\hat{\mathbf{g}}$ has the same distribution as $\mathbf{g}$, i.e., $\hat{\mathbf{g}} \sim \mathcal{N}_d(\mathbf{0}, \mathbf{I})$. Therefore, we get

$$\begin{aligned} \mathbb{E}[\|\mathbf{g}^\top \mathbf{M}\|] &= \mathbb{E}[\|\mathbf{V}\mathbf{\Sigma}^\top \hat{\mathbf{g}}\|] \\ &= \mathbb{E}[\|\mathbf{\Sigma}^\top \hat{\mathbf{g}}\|] \end{aligned}$$

where the second equality follows from the orthogonality of $\mathbf{V}$. Now let us denote the diagonal entries of $\mathbf{\Sigma}$ by $\sigma_1, \sigma_2, \dots, \sigma_r$ (where r is the rank of $\mathbf{M}$) and the i -th coordinate of $\hat{\mathbf{g}}$ by $\hat{g}_i$. Then we have

$$\begin{aligned} \mathbb{E}[\|\mathbf{g}^\top \mathbf{M}\|] &= \mathbb{E}\left[\sqrt{\sum_{i=1}^r \sigma_i^2 \hat{g}_i^2}\right] \\ &= \sqrt{\sum_{j=1}^r \sigma_j^2} \left(\mathbb{E}\left[\sqrt{\sum_{i=1}^r \frac{\sigma_i^2}{\sum_{j=1}^r \sigma_j^2} \hat{g}_i^2}\right] \right) \\ &\geq \sqrt{\sum_{j=1}^r \sigma_j^2} \left(\mathbb{E}\left[\sum_{i=1}^r \frac{\sigma_i^2}{\sum_{j=1}^r \sigma_j^2} \sqrt{\hat{g}_i^2}\right] \right) \\ &= \sqrt{\sum_{j=1}^r \sigma_j^2} \left(\sum_{i=1}^r \frac{\sigma_i^2}{\sum_{j=1}^r \sigma_j^2} \mathbb{E}[\|\hat{g}_i\|] \right) \end{aligned}$$

where the inequality follows from the concavity of $\sqrt{\cdot}$. Finally, as $\hat{g}_i \sim \mathcal{N}(0, 1)$, we know $\mathbb{E}[\|\hat{g}_i\|] = \sqrt{2/\pi}$ and we obtain

$$\begin{aligned} \mathbb{E}[\|\mathbf{g}^\top \mathbf{M}\|] &\geq \sqrt{\frac{2}{\pi}} \sqrt{\sum_{j=1}^r \sigma_j^2} \left(\sum_{i=1}^r \frac{\sigma_i^2}{\sum_{j=1}^r \sigma_j^2} \right) \\ &= \sqrt{\frac{2}{\pi}} \sqrt{\sum_{j=1}^r \sigma_j^2} \\ &= \sqrt{\frac{2}{\pi}} \|\mathbf{M}\|_F. \end{aligned}$$

This completes the proof of the Lemma. $\square$

C ANALYSIS OF MUON WITH NESTEROV'S MOMENTUM AND INEXACT POLAR DECOMPOSITION

Theorem C.1. Under Assumptions A.1, A.2, A.3, and D.2, the update rule of Muon satisfies

$$\begin{aligned} f(\mathbf{X}_{k+1}) &\leq f(\mathbf{X}_k) - \eta \|\mathbf{M}_k\|_* (1 - \Delta_k) + \eta \|\nabla f(\mathbf{X}_k) - \mathbf{M}_k\|_* (\sqrt{d} + \Delta_k) \\ &\quad + \frac{\eta^2 L}{2} (\sqrt{d} + \Delta_k)^2. \end{aligned}$$

Proof. Using the L -smoothness property of f , we obtain

$$\begin{aligned} f(\mathbf{X}_{k+1}) &\leq f(\mathbf{X}_k) + \langle \nabla f(\mathbf{X}_k), \mathbf{X}_{k+1} - \mathbf{X}_k \rangle + \frac{L}{2} \|\mathbf{X}_{k+1} - \mathbf{X}_k\|_F^2 \\ &= f(\mathbf{X}_k) - \eta \langle \nabla f(\mathbf{X}_k), T(\mathbf{M}_k) \rangle + \frac{\eta^2 L}{2} \|T(\mathbf{M}_k)\|_F^2 \\ &= f(\mathbf{X}_k) - \eta \langle \mathbf{M}_k, T(\mathbf{M}_k) \rangle - \eta \langle \nabla f(\mathbf{X}_k) - \mathbf{M}_k, T(\mathbf{M}_k) \rangle + \frac{\eta^2 L}{2} \|T(\mathbf{M}_k)\|_F^2. \end{aligned}$$

For further analysis, let us consider the singular value decomposition (SVD) of $\mathbf{M}_k = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{W}_k^\top$. Here, we will analyze the terms separately to obtain

1. Term I:

$$\begin{aligned} \langle \mathbf{M}_k, T(\mathbf{M}_k) \rangle &= \langle \mathbf{M}_k, O(\mathbf{M}_k) \rangle + \langle \mathbf{M}_k, T(\mathbf{M}_k) - O(\mathbf{M}_k) \rangle \\ &= \langle \mathbf{M}_k, \mathbf{U}_k \mathbf{W}_k^\top \rangle + \langle \mathbf{M}_k, T(\mathbf{M}_k) - O(\mathbf{M}_k) \rangle \\ &= \|\mathbf{M}_k\|_* + \langle \mathbf{M}_k, T(\mathbf{M}_k) - O(\mathbf{M}_k) \rangle \\ &\geq \|\mathbf{M}_k\|_* - \|\mathbf{M}_k\|_F \|T(\mathbf{M}_k) - O(\mathbf{M}_k)\|_F \\ &\geq \|\mathbf{M}_k\|_* - \|\mathbf{M}_k\|_* \|T(\mathbf{M}_k) - O(\mathbf{M}_k)\|_F \\ &\geq \|\mathbf{M}_k\|_* - \Delta_k \|\mathbf{M}_k\|_* \\ &= \|\mathbf{M}_k\|_* (1 - \Delta_k) \end{aligned}$$

where the second-to-last inequality follows from the order of the norm, i.e., $\|\mathbf{M}_k\|_F \leq \|\mathbf{M}_k\|_*$.

2. Term II:

$$-\langle \nabla f(\mathbf{X}_k) - \mathbf{M}_k, T(\mathbf{M}_k) \rangle \leq \|\nabla f(\mathbf{X}_k) - \mathbf{M}_k\|_* \|T(\mathbf{M}_k)\|_F.$$

Note that $\|T(\mathbf{M}_k)\|_F \leq \|O(\mathbf{M}_k)\|_F + \|T(\mathbf{M}_k) - O(\mathbf{M}_k)\|_F = \sqrt{d} + \|T(\mathbf{M}_k) - O(\mathbf{M}_k)\|_F \leq \sqrt{d} + \Delta_k$. Thus, we obtain

$$-\langle \nabla f(\mathbf{X}_k) - \mathbf{M}_k, T(\mathbf{M}_k) \rangle \leq \|\nabla f(\mathbf{X}_k) - \mathbf{M}_k\|_* (\sqrt{d} + \Delta_k).$$

Hence, collecting all the terms, we have

$$\begin{aligned} f(\mathbf{X}_{k+1}) &\leq f(\mathbf{X}_k) - \eta \|\mathbf{M}_k\|_* (1 - \Delta_k) + \eta \|\nabla f(\mathbf{X}_k) - \mathbf{M}_k\|_* (\sqrt{d} + \Delta_k) \\ &\quad + \frac{\eta^2 L}{2} (\sqrt{d} + \Delta_k)^2. \end{aligned}$$

This completes the proof of this Theorem. $\square$

Theorem C.2. Under Assumptions A.1, A.2, A.3, and D.2, the update rule of Muon satisfies

$$\eta(1 - \Delta) \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_*] \leq f(\mathbf{X}_0) - f_* + \eta \left(1 + \sqrt{d}\right) \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k) - \mathbf{M}_k\|_*] + \frac{\eta^2 L K}{2} \left(\sqrt{d} + \Delta\right)^2.$$

when $\Delta_k \leq \Delta < 1$ for all $0 \leq k \leq K - 1$.

Proof. We start from Theorem C.1 to obtain

$$f(\mathbf{X}_{k+1}) \leq f(\mathbf{X}_k) - \eta \|\mathbf{M}_k\|_* (1 - \Delta_k) + \eta \|\nabla f(\mathbf{X}_k) - \mathbf{M}_k\|_* \left(\sqrt{d} + \Delta_k\right) + \frac{\eta^2 L}{2} \left(\sqrt{d} + \Delta_k\right)^2.$$

Using the triangle inequality, we have $\|\mathbf{M}_k\|_* \geq \|\nabla f(\mathbf{X}_k)\|_* - \|\nabla f(\mathbf{X}_k) - \mathbf{M}_k\|_*$. Therefore, we have

$$\begin{aligned} f(\mathbf{X}_{k+1}) &\leq f(\mathbf{X}_k) - \eta \|\nabla f(\mathbf{X}_k)\|_* (1 - \Delta_k) + \eta \|\nabla f(\mathbf{X}_k) - \mathbf{M}_k\|_* (1 - \Delta_k) \\ &\quad + \eta \|\nabla f(\mathbf{X}_k) - \mathbf{M}_k\|_* \left(\sqrt{d} + \Delta_k\right) + \frac{\eta^2 L}{2} \left(\sqrt{d} + \Delta_k\right)^2 \\ &= f(\mathbf{X}_k) - \eta \|\nabla f(\mathbf{X}_k)\|_* (1 - \Delta_k) + \eta \|\nabla f(\mathbf{X}_k) - \mathbf{M}_k\|_* \left(1 + \sqrt{d}\right) \\ &\quad + \frac{\eta^2 L}{2} \left(\sqrt{d} + \Delta_k\right)^2. \end{aligned}$$

Then, rearranging the terms, we have

$$\eta(1 - \Delta_k) \|\nabla f(\mathbf{X}_k)\|_* \leq f(\mathbf{X}_k) - f(\mathbf{X}_{k+1}) + \eta \|\nabla f(\mathbf{X}_k) - \mathbf{M}_k\|_* \left(1 + \sqrt{d}\right) + \frac{\eta^2 L}{2} \left(\sqrt{d} + \Delta_k\right)^2.$$

Here, we assume $\Delta_k \leq \Delta$ for all $0 \leq k \leq K - 1$. Then we have

$$\eta(1 - \Delta) \|\nabla f(\mathbf{X}_k)\|_* \leq f(\mathbf{X}_k) - f(\mathbf{X}_{k+1}) + \eta \|\nabla f(\mathbf{X}_k) - \mathbf{M}_k\|_* \left(1 + \sqrt{d}\right) + \frac{\eta^2 L}{2} \left(\sqrt{d} + \Delta\right)^2.$$

Now we take the total expectation and sum up the terms for $k = 0, 1, \dots, K - 1$ to get

$$\begin{aligned} \eta(1 - \Delta) \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_*] &\leq f(\mathbf{X}_0) - \mathbb{E} [f(\mathbf{X}_K)] \\ &\quad + \eta \left(1 + \sqrt{d}\right) \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k) - \mathbf{M}_k\|_*] \\ &\quad + \frac{\eta^2 L K}{2} \left(\sqrt{d} + \Delta\right)^2 \\ &\leq f(\mathbf{X}_0) - f_* + \eta \left(1 + \sqrt{d}\right) \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k) - \mathbf{M}_k\|_*] \\ &\quad + \frac{\eta^2 L K}{2} \left(\sqrt{d} + \Delta\right)^2. \end{aligned}$$

The last line follows from $\mathbb{E} [f(\mathbf{X}_K)] \geq f_*$. This completes the proof of this Theorem.

Lemma C.3. Let $0 \leq \beta < 1$ and suppose that Assumptions A.1, A.2, A.3 hold. Define

$$\mathbf{S}_k := \mathbf{M}_k - \nabla f(\mathbf{X}_k).$$

Then, for any choice of $\varepsilon \in (0, \beta^{-2} - 1)$ (so that $(1 + \varepsilon)\beta^2 < 1$) and any $K \geq 0$, we have

$$\begin{aligned} \sum_{k=0}^{K-1} \mathbb{E} [\|\mathbf{S}_k\|_*^2] &\leq \frac{2\beta^2}{(1-\lambda)} \mathbb{E} [\|\mathbf{C}_0\|_*^2] \\ &\quad + \frac{4\beta^2(1+1/\varepsilon)}{1-\lambda} \sum_{j=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_j)\|_*^2] \\ &\quad + \frac{4\beta^2(1+1/\varepsilon)}{1-\lambda} d\sigma^2 + 2d\sigma^2. \end{aligned}$$

where

$$\lambda := (1 + \varepsilon)\beta^2 \in (0, 1)$$

□

Proof. For simplicity, let us define the stochastic gradient error and the drift due to momentum given by

$$\begin{aligned} \mathbf{G}_k &:= \nabla f(\mathbf{X}_k) + \boldsymbol{\xi}_k, \\ \mathbf{S}_k &:= \mathbf{M}_k - \nabla f(\mathbf{X}_k). \end{aligned}$$

Thus, we have $\mathbf{S}_k = \mathbf{M}_k - \nabla f(\mathbf{X}_k) = \beta\mathbf{C}_k + \mathbf{G}_k - \nabla f(\mathbf{X}_k) = \beta\mathbf{C}_k + \boldsymbol{\xi}_k$ which implies

$$\begin{aligned} \mathbb{E} [\|\mathbf{S}_k\|_*^2] &\leq 2\beta^2 \mathbb{E} [\|\mathbf{C}_k\|_*^2] + 2\mathbb{E} [\|\boldsymbol{\xi}_k\|_*^2] \\ &\leq 2\beta^2 \mathbb{E} [\|\mathbf{C}_k\|_*^2] + 2d\mathbb{E} [\|\boldsymbol{\xi}_k\|_F^2] \\ &\leq 2\beta^2 \mathbb{E} [\|\mathbf{C}_k\|_*^2] + 2d\sigma^2. \end{aligned} \tag{18}$$

Let $\varepsilon > 0$. Using Young's Inequality, we get

$$\begin{aligned} \|\mathbf{C}_k\|_*^2 &= \|\beta\mathbf{C}_{k-1} + \mathbf{G}_k\|_*^2 \\ &\leq (1 + \varepsilon)\beta^2 \|\mathbf{C}_{k-1}\|_*^2 + \left(1 + \frac{1}{\varepsilon}\right) \|\mathbf{G}_k\|^2 \\ &= (1 + \varepsilon)\beta^2 \|\mathbf{C}_{k-1}\|_*^2 + \left(1 + \frac{1}{\varepsilon}\right) \|\nabla f(\mathbf{X}_k) + \boldsymbol{\xi}_k\|_*^2. \end{aligned}$$

Another use of Young's Inequality gives us

$$\begin{aligned} \mathbb{E} [\|\mathbf{C}_k\|_*^2] &\leq (1 + \varepsilon)\beta^2 \mathbb{E} [\|\mathbf{C}_{k-1}\|_*^2] \\ &\quad + 2\left(1 + \frac{1}{\varepsilon}\right) \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_*^2] + 2\left(1 + \frac{1}{\varepsilon}\right) \mathbb{E} [\|\boldsymbol{\xi}_k\|_*^2] \\ &\leq (1 + \varepsilon)\beta^2 \mathbb{E} [\|\mathbf{C}_{k-1}\|_*^2] \\ &\quad + 2\left(1 + \frac{1}{\varepsilon}\right) \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_*^2] + 2d\left(1 + \frac{1}{\varepsilon}\right) \sigma^2. \end{aligned} \tag{19}$$

Thus

$$\mathbb{E} [\|\mathbf{C}_k\|_*^2] \leq \lambda \mathbb{E} [\|\mathbf{C}_{k-1}\|_*^2] + D_k + C, \tag{20}$$

where we introduced

$$\begin{aligned} \lambda &:= (1 + \varepsilon)\beta^2, \\ D_k &:= 2\left(1 + \frac{1}{\varepsilon}\right) \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_*^2], \\ C &:= 2\left(1 + \frac{1}{\varepsilon}\right) d\sigma^2. \end{aligned}$$

Since $\beta \in [0, 1)$, we can choose $\varepsilon \in (0, \beta^{-2} - 1)$, so that $0 < \lambda = (1 + \varepsilon)\beta^2 < 1$. We now unroll the linear recursion equation 20. By induction,

$$\mathbb{E} [\|C_k\|_*^2] \leq \lambda^k \mathbb{E} [\|C_0\|_*^2] + \sum_{j=1}^k \lambda^{k-j} (D_j + C).$$

Summing over $k = 0, \dots, K-1$, we have

$$\sum_{k=0}^{K-1} \mathbb{E} [\|C_k\|_*^2] \leq \mathbb{E} [\|C_0\|_*^2] \sum_{k=0}^{K-1} \lambda^k + \sum_{k=0}^{K-1} \sum_{j=1}^k \lambda^{k-j} (D_j + C). \quad (21)$$

For the first term, we have the geometric bound. For the double sum, we swap the order of summation:

$$\begin{aligned} \sum_{k=0}^{K-1} \sum_{j=1}^k \lambda^{k-j} (D_j + C) &= \sum_{j=1}^{K-1} (D_j + C) \sum_{k=j}^{K-1} \lambda^{k-j} \\ &\leq \frac{1}{1-\lambda} \sum_{j=1}^{K-1} (D_j + C) \\ &= \frac{1}{1-\lambda} \left(\sum_{j=1}^{K-1} D_j + C(K-1) \right). \end{aligned}$$

Plugging into equation 21, we obtain

$$\sum_{k=0}^{K-1} \mathbb{E} [\|C_k\|_*^2] \leq \frac{\mathbb{E} [\|C_0\|_*^2]}{(1-\lambda)} + \frac{1}{(1-\lambda)} \sum_{j=1}^{K-1} D_j + \frac{C(K-1)}{1-\lambda}. \quad (22)$$

Thus, we obtain

$$\begin{aligned} \sum_{k=0}^{K-1} \mathbb{E} [\|S_k\|_*^2] &= 2\beta^2 \sum_{k=0}^{K-1} \mathbb{E} [\|C_k\|_*^2] + 2(K-1)d\sigma^2 \\ &\leq \frac{2\beta^2}{(1-\lambda)} \mathbb{E} [\|C_0\|_*^2] \\ &\quad + \frac{4\beta^2(1+1/\varepsilon)}{1-\lambda} \sum_{j=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_j)\|_*^2] \\ &\quad + \frac{4\beta^2(1+1/\varepsilon)}{1-\lambda} (K-1)d\sigma^2 + 2(K-1)d\sigma^2. \end{aligned}$$

This completes the proof of the Lemma. $\square$

Theorem C.4. Let $0 \leq \beta < 1$ and suppose that Assumptions A.1, A.3 hold with $\Delta_k \leq \Delta < 1$. Then MUON with Nesterov's momentum satisfies

$$\begin{aligned} \min_{0 \leq k \leq K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_*] &\leq \frac{f(\mathbf{X}_0) - f_*}{\eta \rho K} + \frac{2\beta^2 (1 + \sqrt{d})}{(1 - \lambda) \rho K} \mathbb{E} [\|\mathbf{C}_0\|_*^2] \\ &\quad + \frac{\eta L}{2\rho} (\sqrt{d} + \Delta)^2 \\ &\quad + \frac{2 (1 + \sqrt{d}) d \sigma^2}{\rho} \left(1 + \frac{2\beta^2 (1 + 1/\varepsilon)}{1 - \lambda}\right) \end{aligned}$$

where β is such that

$$\rho := \left[(1 - \Delta) - (1 + \sqrt{d}) \frac{4\beta^2 (1 + 1/\varepsilon)}{1 - \lambda} \right] > 0.$$

Proof. Thus, we have

$$\begin{aligned} \eta \left[(1 - \Delta) - \frac{4\beta^2 (1 + 1/\varepsilon) (1 + \sqrt{d})}{1 - \lambda} \right] \sum_{k=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_*] &\leq f(\mathbf{X}_0) - f_* \\ &\quad + \frac{2\eta\beta^2 (1 + \sqrt{d})}{(1 - \lambda)} \mathbb{E} [\|\mathbf{C}_0\|_*^2] \\ &\quad + \frac{\eta^2 L K}{2} (\sqrt{d} + \Delta)^2 \\ &\quad + 2\eta (1 + \sqrt{d}) (K - 1) d \sigma^2 \left(1 + \frac{2\beta^2 (1 + 1/\varepsilon)}{1 - \lambda}\right) \end{aligned}$$

Here we can choose β small enough such that

$$\rho := \left[(1 - \Delta) - (1 + \sqrt{d}) \frac{4\beta^2 (1 + 1/\varepsilon)}{1 - \lambda} \right] > 0.$$

Hence, we have

$$\begin{aligned} \min_{0 \leq k \leq K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_k)\|_*] &\leq \frac{f(\mathbf{X}_0) - f_*}{\eta \rho K} \\ &\quad + \frac{2\beta^2 (1 + \sqrt{d})}{(1 - \lambda) \rho K} \mathbb{E} [\|\mathbf{C}_0\|_*^2] \\ &\quad + \frac{\eta L}{2\rho} (\sqrt{d} + \Delta)^2 \\ &\quad + \frac{2 (1 + \sqrt{d}) d \sigma^2}{\rho} \left(1 + \frac{2\beta^2 (1 + 1/\varepsilon)}{1 - \lambda}\right) \end{aligned}$$

□

Lemma C.5 (Uniform nonlinear Young inequality for $1 < \alpha \leq 2$). Let $\alpha \in (1, 2]$ and $\varepsilon > 0$. Define

$$C(\varepsilon) := 16\left(1 + \frac{1}{\varepsilon}\right).$$

Then, for any norm $\|\cdot\|$ on a vector space and any $\mathbf{A}, \mathbf{B}$,

$$\|\mathbf{A} + \mathbf{B}\|^\alpha \leq (1 + \varepsilon)\|\mathbf{A}\|^\alpha + C(\varepsilon)\|\mathbf{B}\|^\alpha. \quad (23)$$

This bound is uniform over all $\alpha \in (1, 2]$.

Proof. Let $x := \|\mathbf{A}\| \geq 0$, $y := \|\mathbf{B}\| \geq 0$. If $x = 0$ the inequality is trivial since $(x + y)^\alpha = y^\alpha \leq C(\varepsilon)y^\alpha$ for $C(\varepsilon) \geq 1$. Suppose $x > 0$ and set $t := y/x \geq 0$. Then

$$(x + y)^\alpha = x^\alpha(1 + t)^\alpha \quad \text{and} \quad x^\alpha + C(\varepsilon)y^\alpha = x^\alpha(1 + C(\varepsilon)t^\alpha).$$

Thus it suffices to show

$$(1 + t)^\alpha \leq (1 + \varepsilon) + C(\varepsilon)t^\alpha \quad \text{for all } t \geq 0, \alpha \in (1, 2].$$

We treat $t \in [0, 1]$ and $t \geq 1$ separately.

Case 1: $t \geq 1$. We have

$$(1 + t)^\alpha \leq (2t)^\alpha = 2^\alpha t^\alpha \leq 4t^\alpha,$$

since $\alpha \leq 2$. Therefore

$$(1 + t)^\alpha - (1 + \varepsilon) \leq 4t^\alpha \leq C(\varepsilon)t^\alpha$$

as soon as $C(\varepsilon) \geq 4$, which holds for $C(\varepsilon) = 16(1 + 1/\varepsilon)$.

Case 2: $0 \leq t \leq 1$. Using the mean value theorem on $g(u) = u^\alpha$ between $u = 1$ and $u = 1 + t$, there exists $\xi \in [1, 1 + t] \subseteq [1, 2]$ such that

$$(1 + t)^\alpha - 1 = \alpha\xi^{\alpha-1}t \leq \alpha 2^{\alpha-1}t \leq 4t,$$

since $\alpha \leq 2$ and $2^{\alpha-1} \leq 2$. Hence

$$(1 + t)^\alpha \leq 1 + 4t.$$

If $0 \leq t \leq \varepsilon/4$, then $1 + 4t \leq 1 + \varepsilon$ and

$$(1 + t)^\alpha - (1 + \varepsilon) \leq 0 \leq C(\varepsilon)t^\alpha.$$

If $t \in [\varepsilon/4, 1]$, then

$$(1 + t)^\alpha - (1 + \varepsilon) \leq 4t \leq 4t^\alpha \left(\frac{4}{\varepsilon}\right)^{\alpha-1},$$

because $t^{1-\alpha}$ is decreasing in t for $\alpha > 1$ and $t \geq \varepsilon/4$. Thus

$$\frac{(1 + t)^\alpha - (1 + \varepsilon)}{t^\alpha} \leq 4 \left(\frac{4}{\varepsilon}\right)^{\alpha-1} \leq 4 \left(\frac{4}{\varepsilon}\right)^1 = \frac{16}{\varepsilon},$$

since $\alpha - 1 \leq 1$. Therefore

$$(1 + t)^\alpha - (1 + \varepsilon) \leq \frac{16}{\varepsilon} t^\alpha \leq C(\varepsilon)t^\alpha,$$

because $C(\varepsilon) = 16(1 + 1/\varepsilon) \geq 16/\varepsilon$.

Combining both cases shows that

$$(1 + t)^\alpha \leq (1 + \varepsilon) + C(\varepsilon)t^\alpha \quad \text{for all } t \geq 0, \alpha \in (1, 2],$$

which is exactly equation 23. $\square$

Lemma C.6 (Nuclear-norm bound with α -moment noise). Let $0 \leq \beta < 1$ and suppose that Assumptions A.1, A.2, and A.4 hold for some $\alpha \in (1, 2]$. Define

$$\mathbf{S}_k := \mathbf{M}_k - \nabla f(\mathbf{X}_k).$$

Then, for any $\varepsilon > 0$ such that

$$\lambda := (1 + \varepsilon) \beta^\alpha \in (0, 1) \quad (\text{i.e. } \varepsilon \in (0, \beta^{-\alpha} - 1)),$$

and any integer $K \geq 1$, we have

$$\begin{aligned} \sum_{k=0}^{K-1} \mathbb{E} [\|\mathbf{S}_k\|_*^\alpha] &\leq \frac{(1 + \varepsilon) \beta^\alpha}{1 - \lambda} \mathbb{E} [\|\mathbf{C}_0\|_*^\alpha] \\ &\quad + \frac{(1 + \varepsilon)^2 \beta^\alpha C(\varepsilon)}{1 - \lambda} \sum_{j=0}^{K-1} \mathbb{E} [\|\nabla f(\mathbf{X}_j)\|_*^\alpha] \\ &\quad + \left(\frac{(1 + \varepsilon) \beta^\alpha C(\varepsilon)^2}{1 - \lambda} + C(\varepsilon) \right) K d^{\alpha/2} \sigma^\alpha, \end{aligned}$$

where $C(\varepsilon) = 16(1 + 1/\varepsilon)$ is as in Lemma C.5.

Proof. We begin with the stochastic gradient decomposition

$$\mathbf{G}_k = \nabla f(\mathbf{X}_k) + \boldsymbol{\xi}_k,$$

with $\mathbb{E} [\boldsymbol{\xi}_k \mid \mathbf{X}_k] = 0$ and $\mathbb{E} [\|\boldsymbol{\xi}_k\|_F^\alpha \mid \mathbf{X}_k] \leq \sigma^\alpha$ by Assumption A.4. For convenience, recall

$$\mathbf{S}_k := \mathbf{M}_k - \nabla f(\mathbf{X}_k), \quad \mathbf{M}_k = \beta \mathbf{C}_k + \mathbf{G}_k.$$

Step 1: Expressing $\mathbf{S}_k$ via $\mathbf{C}_k$ and $\boldsymbol{\xi}_k$. From $\mathbf{M}_k = \beta \mathbf{C}_k + \mathbf{G}_k$ and $\mathbf{G}_k = \nabla f(\mathbf{X}_k) + \boldsymbol{\xi}_k$, we obtain

$$\begin{aligned} \mathbf{S}_k &= \mathbf{M}_k - \nabla f(\mathbf{X}_k) \\ &= \beta \mathbf{C}_k + \mathbf{G}_k - \nabla f(\mathbf{X}_k) \\ &= \beta \mathbf{C}_k + \boldsymbol{\xi}_k. \end{aligned}$$

Applying Lemma C.5 with the nuclear norm, $\mathbf{A} = \beta \mathbf{C}_k$, $\mathbf{B} = \boldsymbol{\xi}_k$, we get

$$\begin{aligned} \|\mathbf{S}_k\|_*^\alpha &= \|\beta \mathbf{C}_k + \boldsymbol{\xi}_k\|_*^\alpha \\ &\leq (1 + \varepsilon) \beta^\alpha \|\mathbf{C}_k\|_*^\alpha + C(\varepsilon) \|\boldsymbol{\xi}_k\|_*^\alpha. \end{aligned} \tag{24}$$

Taking expectations and using $\|\boldsymbol{\xi}_k\|_* \leq \sqrt{d} \|\boldsymbol{\xi}_k\|_F$ and $\mathbb{E} [\|\boldsymbol{\xi}_k\|_F^\alpha] \leq \sigma^\alpha$, we obtain

$$\mathbb{E} [\|\mathbf{S}_k\|_*^\alpha] \leq (1 + \varepsilon) \beta^\alpha \mathbb{E} [\|\mathbf{C}_k\|_*^\alpha] + C(\varepsilon) d^{\alpha/2} \sigma^\alpha. \tag{25}$$

Step 2: Recursion for $\mathbb{E} [\|\mathbf{C}_k\|_*^\alpha]$. From the momentum recursion,

$$\mathbf{C}_k = \beta \mathbf{C}_{k-1} + \mathbf{G}_k = \beta \mathbf{C}_{k-1} + \nabla f(\mathbf{X}_k) + \boldsymbol{\xi}_k,$$

we apply Lemma C.5 to $\beta \mathbf{C}_{k-1} + \mathbf{G}_k$:

$$\begin{aligned} \|\mathbf{C}_k\|_*^\alpha &= \|\beta \mathbf{C}_{k-1} + \mathbf{G}_k\|_*^\alpha \\ &\leq (1 + \varepsilon) \beta^\alpha \|\mathbf{C}_{k-1}\|_*^\alpha + C(\varepsilon) \|\mathbf{G}_k\|_*^\alpha. \end{aligned} \tag{26}$$

Next, apply Lemma C.5 again to $\nabla f(\mathbf{X}_k) + \boldsymbol{\xi}_k$:

$$\begin{aligned} \|\mathbf{G}_k\|_*^\alpha &= \|\nabla f(\mathbf{X}_k) + \boldsymbol{\xi}_k\|_*^\alpha \\ &\leq (1 + \varepsilon) \|\nabla f(\mathbf{X}_k)\|_*^\alpha + C(\varepsilon) \|\boldsymbol{\xi}_k\|_*^\alpha. \end{aligned} \tag{27}$$

Combining equation 26 and equation 27 yields

$$\begin{aligned} \|\mathbf{C}_k\|_*^\alpha &\leq (1 + \varepsilon) \beta^\alpha \|\mathbf{C}_{k-1}\|_*^\alpha \\ &\quad + C(\varepsilon) (1 + \varepsilon) \|\nabla f(\mathbf{X}_k)\|_*^\alpha + C(\varepsilon)^2 \|\boldsymbol{\xi}_k\|_*^\alpha. \end{aligned} \tag{28}$$

Taking expectations and using $\mathbb{E}[\|\xi_k\|_*^\alpha] \leq d^{\alpha/2}\sigma^\alpha$, we obtain

$$\mathbb{E}[\|C_k\|_*^\alpha] \leq \lambda \mathbb{E}[\|C_{k-1}\|_*^\alpha] + D_k + C_0, \quad (29)$$

where

$$\begin{aligned} \lambda &:= (1 + \varepsilon)\beta^\alpha, \\ D_k &:= C(\varepsilon)(1 + \varepsilon)\mathbb{E}[\|\nabla f(\mathbf{X}_k)\|_*^\alpha], \\ C_0 &:= C(\varepsilon)^2 d^{\alpha/2}\sigma^\alpha. \end{aligned}$$

By assumption on ε , we have $0 < \lambda < 1$.

Unrolling equation 29 by induction gives

$$\mathbb{E}[\|C_k\|_*^\alpha] \leq \lambda^k \mathbb{E}[\|C_0\|_*^\alpha] + \sum_{j=1}^k \lambda^{k-j} (D_j + C_0).$$

Summing over $k = 0, \dots, K-1$ yields

$$\sum_{k=0}^{K-1} \mathbb{E}[\|C_k\|_*^\alpha] \leq \mathbb{E}[\|C_0\|_*^\alpha] \sum_{k=0}^{K-1} \lambda^k + \sum_{k=0}^{K-1} \sum_{j=1}^k \lambda^{k-j} (D_j + C_0). \quad (30)$$

Since $\sum_{k=0}^{K-1} \lambda^k \leq 1/(1-\lambda)$, and swapping the order of summation in the double sum,

$$\begin{aligned} \sum_{k=0}^{K-1} \sum_{j=1}^k \lambda^{k-j} (D_j + C_0) &= \sum_{j=1}^{K-1} (D_j + C_0) \sum_{k=j}^{K-1} \lambda^{k-j} \\ &\leq \frac{1}{1-\lambda} \sum_{j=1}^{K-1} (D_j + C_0), \end{aligned}$$

we obtain

$$\sum_{k=0}^{K-1} \mathbb{E}[\|C_k\|_*^\alpha] \leq \frac{1}{1-\lambda} \mathbb{E}[\|C_0\|_*^\alpha] + \frac{1}{1-\lambda} \sum_{j=1}^{K-1} D_j + \frac{C_0(K-1)}{1-\lambda}. \quad (31)$$

Step 3: Final bound for S_k . Summing equation 25 over $k = 0, \dots, K-1$ and using equation 31, we obtain

$$\begin{aligned} \sum_{k=0}^{K-1} \mathbb{E}[\|S_k\|_*^\alpha] &\leq (1 + \varepsilon)\beta^\alpha \sum_{k=0}^{K-1} \mathbb{E}[\|C_k\|_*^\alpha] + KC(\varepsilon)d^{\alpha/2}\sigma^\alpha \\ &\leq \frac{(1 + \varepsilon)\beta^\alpha}{1-\lambda} \mathbb{E}[\|C_0\|_*^\alpha] \\ &\quad + \frac{(1 + \varepsilon)\beta^\alpha}{1-\lambda} \sum_{j=1}^{K-1} D_j \\ &\quad + \frac{(1 + \varepsilon)\beta^\alpha C_0}{1-\lambda} (K-1) + KC(\varepsilon)d^{\alpha/2}\sigma^\alpha. \end{aligned}$$

Extending $\sum_{j=1}^{K-1}$ to $\sum_{j=0}^{K-1}$ (only adding a nonnegative term) and substituting

$$D_j = C(\varepsilon)(1 + \varepsilon)\mathbb{E}[\|\nabla f(\mathbf{X}_j)\|_*^\alpha], \quad C_0 = C(\varepsilon)^2 d^{\alpha/2}\sigma^\alpha,$$

gives

$$\begin{aligned} \sum_{k=0}^{K-1} \mathbb{E}[\|S_k\|_*^\alpha] &\leq \frac{(1 + \varepsilon)\beta^\alpha}{1-\lambda} \mathbb{E}[\|C_0\|_*^\alpha] \\ &\quad + \frac{(1 + \varepsilon)^2 \beta^\alpha C(\varepsilon)}{1-\lambda} \sum_{j=0}^{K-1} \mathbb{E}[\|\nabla f(\mathbf{X}_j)\|_*^\alpha] \\ &\quad + \left(\frac{(1 + \varepsilon)\beta^\alpha C(\varepsilon)^2}{1-\lambda} + C(\varepsilon) \right) K d^{\alpha/2} \sigma^\alpha. \end{aligned}$$

This is the desired bound. $\square$

D PREVIOUS ANALYSIS

Lemma D.1. Under the Assumptions A.1, A.2, and A.3, the update rule of Muon with Nesterov's momentum satisfies

$$\begin{aligned} \mathbb{E} [\| \mathbf{X}_{k+1} - \mathbf{X}_* \|_F^2 \mid \mathbf{X}_k] &\leq \left(1 - \eta\mu + \frac{\eta}{\alpha}\right) \|\mathbf{X}_k - \mathbf{X}_*\|_F^2 - 2\eta(1 - 2\eta L)(f(\mathbf{X}_k) - f(\mathbf{X}_*)) \\ &\quad + \eta(\alpha + 2\eta)\mathbb{E} [\|T(\mathbf{M}_k) - \mathbf{G}_k\|_F^2 \mid \mathbf{X}_k] + 2\eta^2\sigma^2. \end{aligned}$$

for any $\alpha > 0$.

Proof. From the update rule of Muon, we have

$$\begin{aligned} \|\mathbf{X}_{k+1} - \mathbf{X}_*\|_F^2 &= \|\mathbf{X}_k - \eta T(\mathbf{M}_k) - \mathbf{X}_*\|_F^2 \\ &= \|\mathbf{X}_k - \mathbf{X}_*\|_F^2 - 2\eta\langle T(\mathbf{M}_k), \mathbf{X}_k - \mathbf{X}_* \rangle + \eta^2\|T(\mathbf{M}_k)\|_F^2 \\ &= \|\mathbf{X}_k - \mathbf{X}_*\|_F^2 - 2\eta\langle \mathbf{G}_k, \mathbf{X}_k - \mathbf{X}_* \rangle - 2\eta\langle T(\mathbf{M}_k) - \mathbf{G}_k, \mathbf{X}_k - \mathbf{X}_* \rangle \\ &\quad + \eta^2\|T(\mathbf{M}_k)\|_F^2. \end{aligned}$$

Then, taking the expectation conditioned on $\mathbf{X}_k$, we obtain

$$\begin{aligned} \mathbb{E} [\|\mathbf{X}_{k+1} - \mathbf{X}_*\|_F^2 \mid \mathbf{X}_k] &= \|\mathbf{X}_k - \mathbf{X}_*\|_F^2 - 2\eta\langle \nabla f(\mathbf{X}_k), \mathbf{X}_k - \mathbf{X}_* \rangle \\ &\quad - 2\eta\mathbb{E} [\langle T(\mathbf{M}_k) - \mathbf{G}_k, \mathbf{X}_k - \mathbf{X}_* \rangle \mid \mathbf{X}_k] + \eta^2\mathbb{E} [\|T(\mathbf{M}_k)\|_F^2 \mid \mathbf{X}_k] \\ &\leq \|\mathbf{X}_k - \mathbf{X}_*\|_F^2 - 2\eta(f(\mathbf{X}_k) - f(\mathbf{X}_*)) - \eta\mu\|\mathbf{X}_k - \mathbf{X}_*\|_F^2 \\ &\quad - 2\eta\mathbb{E} [\langle T(\mathbf{M}_k) - \mathbf{G}_k, \mathbf{X}_k - \mathbf{X}_* \rangle \mid \mathbf{X}_k] + \eta^2\mathbb{E} [\|T(\mathbf{M}_k)\|_F^2 \mid \mathbf{X}_k] \\ &= (1 - \eta\mu)\|\mathbf{X}_k - \mathbf{X}_*\|_F^2 - 2\eta(f(\mathbf{X}_k) - f(\mathbf{X}_*)) \\ &\quad - 2\eta\mathbb{E} [\langle T(\mathbf{M}_k) - \mathbf{G}_k, \mathbf{X}_k - \mathbf{X}_* \rangle \mid \mathbf{X}_k] + \eta^2\mathbb{E} [\|T(\mathbf{M}_k)\|_F^2 \mid \mathbf{X}_k]. \end{aligned}$$

Here, the second line follows from using μ -strong monotonicity of f . Note that the third term can be bounded as follows.

$$\begin{aligned} -\eta\langle T(\mathbf{M}_k) - \mathbf{G}_k, \mathbf{X}_k - \mathbf{X}_* \rangle &\leq \eta\|T(\mathbf{M}_k) - \mathbf{G}_k\|_F\|\mathbf{X}_k - \mathbf{X}_*\|_F \\ &\leq \eta\left(\frac{\alpha}{2}\|T(\mathbf{M}_k) - \mathbf{G}_k\|_F^2 + \frac{1}{2\alpha}\|\mathbf{X}_k - \mathbf{X}_*\|_F^2\right). \end{aligned}$$

where the first line follows from Cauchy-Schwarz, while the second line follows from Young's inequality. Here $\alpha > 0$ is a free parameter that will be chosen later. Therefore, we get

$$\begin{aligned} \mathbb{E} [\|\mathbf{X}_{k+1} - \mathbf{X}_*\|_F^2 \mid \mathbf{X}_k] &\leq \left(1 - \eta\mu + \frac{\eta}{\alpha}\right) \|\mathbf{X}_k - \mathbf{X}_*\|_F^2 - 2\eta(f(\mathbf{X}_k) - f(\mathbf{X}_*)) \\ &\quad + \eta\alpha\mathbb{E} [\|T(\mathbf{M}_k) - \mathbf{G}_k\|_F^2 \mid \mathbf{X}_k] + \eta^2\mathbb{E} [\|T(\mathbf{M}_k)\|_F^2 \mid \mathbf{X}_k] \\ &\leq \left(1 - \eta\mu + \frac{\eta}{\alpha}\right) \|\mathbf{X}_k - \mathbf{X}_*\|_F^2 - 2\eta(f(\mathbf{X}_k) - f(\mathbf{X}_*)) \\ &\quad + \eta\alpha\mathbb{E} [\|T(\mathbf{M}_k) - \mathbf{G}_k\|_F^2 \mid \mathbf{X}_k] + 2\eta^2\mathbb{E} [\|T(\mathbf{M}_k) - \mathbf{G}_k\|_F^2 \mid \mathbf{X}_k] \\ &\quad + 2\eta^2\mathbb{E} [\|\mathbf{G}_k\|_F^2 \mid \mathbf{X}_k]. \end{aligned}$$

The last line follows from applying Young's inequality. Then, using the bounded variance assumption, we have

$$\begin{aligned} \mathbb{E} [\|\mathbf{X}_{k+1} - \mathbf{X}_*\|_F^2 \mid \mathbf{X}_k] &\leq \left(1 - \eta\mu + \frac{\eta}{\alpha}\right) \|\mathbf{X}_k - \mathbf{X}_*\|_F^2 - 2\eta(f(\mathbf{X}_k) - f(\mathbf{X}_*)) \\ &\quad + \eta(\alpha + 2\eta)\mathbb{E} [\|T(\mathbf{M}_k) - \mathbf{G}_k\|_F^2 \mid \mathbf{X}_k] + 2\eta^2\|\nabla f(\mathbf{X}_k)\|_F^2 + 2\eta^2\sigma^2 \\ &\leq \left(1 - \eta\mu + \frac{\eta}{\alpha}\right) \|\mathbf{X}_k - \mathbf{X}_*\|_F^2 - 2\eta(f(\mathbf{X}_k) - f(\mathbf{X}_*)) \\ &\quad + \eta(\alpha + 2\eta)\mathbb{E} [\|T(\mathbf{M}_k) - \mathbf{G}_k\|_F^2 \mid \mathbf{X}_k] + 4\eta^2L(f(\mathbf{X}_k) - f(\mathbf{X}_*)) \\ &\quad + 2\eta^2\sigma^2 \\ &= \left(1 - \eta\mu + \frac{\eta}{\alpha}\right) \|\mathbf{X}_k - \mathbf{X}_*\|_F^2 - 2\eta(1 - 2\eta L)(f(\mathbf{X}_k) - f(\mathbf{X}_*)) \\ &\quad + \eta(\alpha + 2\eta)\mathbb{E} [\|T(\mathbf{M}_k) - \mathbf{G}_k\|_F^2 \mid \mathbf{X}_k] + 2\eta^2\sigma^2. \end{aligned}$$

where the second line follows from the Lipschitz property of f . This completes the proof of the lemma. $\square$

Remark: Note that the above lemma does not make any assumptions about the operator T and the matrix M_k . Therefore, the lemma is true for both Nesterov's and Polyak's momentum. Moreover, the lemma also holds for both exact and inexact Polar decomposition of M_k , as we did not make any assumptions about T .

For convenience, let us use $O(M_k)$ to denote the true polar decomposition of M_k . Then we can bound $\|T(M_k) - G_k\|_F^2$ as follows.

$$\begin{aligned} \|T(M_k) - G_k\|_F^2 &= \|T(M_k) - O(M_k) + O(M_k) - M_k + M_k - G_k\|_F^2 \\ &\leq 3\|T(M_k) - O(M_k)\|_F^2 + 3\|O(M_k) - M_k\|_F^2 + 3\|M_k - G_k\|_F^2 \end{aligned}$$

where the first term captures the error from computing the polar decomposition using the Newton-Schultz or PolarExpress method; the second term denotes the change in gradient due to performing the Polar decomposition, whereas the third term denotes the drift due to implementing Nesterov's momentum. Now we discuss the analysis of each of these terms separately.

Analysis of Term I: For the approximation error of using the Newton-Schulz or Polar Express, we can make the following assumption similar to [Shulgin et al. \(2025\)](#).

Assumption D.2. There exists a sequence of $\Delta_k \geq 0$ such that

$$\|T(M_k) - O(M_k)\|_F \leq \Delta_k.$$

Analysis of Term II: Expanding the second term, we get

$$\begin{aligned} \|O(M_k) - M_k\|_F^2 &= \|O(M_k)\|_F^2 + \|M_k\|_F^2 - 2\langle O(M_k), M_k \rangle \\ &= d + \|M_k\|_F^2 - 2\langle O(M_k), M_k \rangle. \end{aligned}$$

The last equality follows from the orthogonality of $O(M_k)$. Suppose the singular value decomposition of M_k is given by $M_k = U_k \Sigma_k W_k^\top$. Then, the inner product of M_k and $O(M_k)$ is lower bounded by

$$\begin{aligned} \langle O(M_k), M_k \rangle &= \langle U_k W_k^\top, M_k \rangle \\ &= \text{tr}(\Sigma_k) \\ &= \|M_k\|_* \\ &\geq \|M_k\|_F. \end{aligned}$$

Thus, by combining the two inequalities mentioned above, we have

$$\begin{aligned} \|O(M_k) - M_k\|_F^2 &= \|O(M_k)\|_F^2 + \|M_k\|_F^2 - 2\langle O(M_k), M_k \rangle \\ &= d + \|M_k\|_F^2 - 2\|M_k\|_F \\ &\leq d + \|M_k\|_F^2. \end{aligned}$$

Analysis of Term III: From the update rule of Muon, note that $\|M_k - G_k\|_F^2 = \beta^2 \|C_k\|_F^2$, and we can expand C_k as follows.

$$\begin{aligned} C_k &= \beta C_{k-1} + G_k \\ &= \beta(\beta C_{k-2} + G_{k-1}) + G_k \\ &= \beta^2 C_{k-2} + \beta G_{k-1} + G_k \\ &\vdots \\ &= \beta^{k+1} C_{-1} + \sum_{j=0}^k \beta^{k-j} G_j. \end{aligned}$$

Then, by taking expectations on both sides, we obtain

$$\begin{aligned}
\mathbb{E} [\|\mathbf{C}_k\|_F^2] &= \mathbb{E} \left[\left\| \sum_{j=0}^k \beta^{k-j} \mathbf{G}_j \right\|_F^2 \right] \\
&\leq (k+1) \sum_{j=0}^k \beta^{2(k-j)} \mathbb{E} [\|\mathbf{G}_j\|_F^2] \\
&\leq (k+1) \sum_{j=0}^k \beta^{2(k-j)} (\|\nabla f(\mathbf{X}_j)\|_F^2 + \sigma^2) \\
&= (k+1) \sum_{j=0}^k \beta^{2(k-j)} \|\nabla f(\mathbf{X}_j)\|_F^2 + (k+1) \sum_{j=0}^k \beta^{2(k-j)} \sigma^2.
\end{aligned}$$

From the update rule of Muon, we have

$$\begin{aligned}
\|\mathbf{C}_{k+1}\|_F^2 &= \|\beta \mathbf{C}_k + \mathbf{G}_{k+1}\|_F^2 \\
&\leq 2\beta^2 \|\mathbf{C}_k\|_F^2 + 2\|\mathbf{G}_{k+1}\|_F^2.
\end{aligned}$$

Then, by taking expectations on both sides, we obtain

$$\begin{aligned}
\mathbb{E} [\|\mathbf{C}_{k+1}\|_F^2] &\leq 2\beta^2 \mathbb{E} [\|\mathbf{C}_k\|_F^2] + 2\mathbb{E} [\|\mathbf{G}_{k+1}\|_F^2] \\
&\leq 2\beta^2 \mathbb{E} [\|\mathbf{C}_k\|_F^2] + 2\mathbb{E} [\|\nabla f(\mathbf{X}_{k+1})\|_F^2] + 2\sigma^2 \\
&\leq 2\beta^2 \mathbb{E} [\|\mathbf{C}_k\|_F^2] + 2L^2 \mathbb{E} [\|\mathbf{X}_{k+1} - \mathbf{X}_*\|_F^2] + 2\sigma^2.
\end{aligned}$$

Thus, by combining it with $\mathbb{E} [\|\mathbf{X}_{k+1} - \mathbf{X}_*\|_F^2]$, we have

$$\begin{aligned}
\mathbb{E} [\|\mathbf{X}_{k+1} - \mathbf{X}_*\|_F^2] + \gamma^2 \mathbb{E} [\|\mathbf{C}_{k+1}\|_F^2] &\leq (1 + 2\gamma^2 L^2) \mathbb{E} [\|\mathbf{X}_{k+1} - \mathbf{X}_*\|_F^2] \\
&\quad + 2\gamma^2 \beta^2 \mathbb{E} [\|\mathbf{C}_k\|_F^2] + 2\gamma^2 \sigma^2
\end{aligned}$$